WORKING PAPER

Evaluating Agricultural Loans and Expenditure Allocation in Nigeria Using Double Machine Learning

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Abstract

Agricultural credit is widely promoted as a policy instrument for advancing food security and agricultural transformation in low- and middle-income countries. Yet evidence on how smallholders allocate the credit they receive remains mixed, particularly in contexts characterized by limited formal financial penetration. This paper examines how agricultural credit affects household expenditure allocation in Nigeria using panel data and a Double/Debiased Machine Learning (DML) framework. We estimate the causal impact of receiving an agricultural loan on farm and nonfarm expenditures and allow treatment effects to vary continuously with farm size. On average, credit increases farm and nonfarm expenditures by 14.6 and 14.3 percent respectively. We also find evidence of heterogeneity: disaggregated outcome analysis identifies credit-induced fertilizer intensification that rises with farm size. The gradient is recovered by DML but absent under a conventional two-way fixed effects benchmark. Less precise aggregate results suggest total farm investment rises with landholdings, while the effect on nonfarm expenditures does not vary systematically with scale. These findings indicate that agricultural credit operates through differentiated channels depending on production capacity, supporting intensification where scale permits and primarily relaxing liquidity constraints among smaller farms. The results highlight the importance of accounting for heterogeneity in the design and evaluation of agricultural finance policies. They also show that nonparametric estimation adds the most value where expenditure responses are nonlinear in scale.

Key Words: agricultural credit, household expenditures, Nigeria, double/debiased machine learning
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1 Introduction

Across low- and middle-income countries, agricultural credit has become a central policy instrument for advancing food security, rural development, and agricultural transformation. Governments and multilateral institutions increasingly rely on subsidized lending, credit guarantees, and risk-sharing facilities to relax liquidity constraints and promote farm investment (FAO, 2019). In principle, expanded access to agricultural finance enables farmers to invest in productivity-enhancing inputs, smooth seasonal cash flows, and manage risk.

In Nigeria, where smallholders dominate the farming population (FAO, 2018) and credit markets remain largely informal, agricultural credit occupies a prominent place in rural development policy. While some programs aimed at boosting credit have been associated with increases in aggregate production (Umeh and Adejo, 2019), credit penetration remains low: only about 5 percent of farming households report receiving loans for agricultural purposes (World Bank Group, 2025a), and most loans originate from informal or semi-formal sources.

This environment raises a persistent empirical question in smallholder agricultural finance: when farmers gain access to credit, does it translate into productive farm expenditures, does it provide liquidity relief for routine consumption expenditures, or both? Understanding how credit is deployed once received (and whether responses vary across farmers with different production endowments) remains central to evaluating credit-based interventions.

We address this question using panel data from Nigeria and a Double/Debiased Machine Learning (DML) framework. We estimate the causal impact of receiving an agricultural loan on household expenditure allocation, distinguishing between farm expenditures (inputs, land, labor, and livestock) and nonfarm expenditures (food and other household consumption). We further examine heterogeneous treatment effects as a continuous function of farm size to test whether credit responses vary systematically across the land distribution.

A large empirical literature examines whether relaxing credit constraints translates into higher agricultural investment and productivity. Early work emphasizes that credit access affects both the level and composition of farm investment under incomplete markets (Carter and Olinto, 2003). More recent experimental and quasi-experimental studies find that credit can increase agricultural investment, but that impacts are highly heterogeneous and depend on farm scale, timing, and complementary institutions such as insurance or savings mechanisms (Karlan et al., 2014; Burke et al., 2019; Fink et al., 2020). In many settings, larger or less constrained farms are better positioned to convert liquidity into productivity-enhancing investments, while smaller farms face structural constraints that limit returns to credit-financed input use. Consistent with this, a parallel literature shows that agricultural and microenterprise credit frequently functions as general household liquidity rather than narrowly targeted investment finance. Borrowers often reallocate funds toward consumption smoothing, risk management, or nonfarm activities, particularly in

environments characterized by income volatility, food insecurity, and missing insurance markets (Angelucci et al., 2015; Banerjee et al., 2015; Meager, 2019). These patterns align with household portfolio optimization under binding constraints, where stabilizing consumption and food security may dominate marginal on-farm investment, especially among smaller or more vulnerable farms.

Our results indicate that receiving an agricultural loan increases both farm and nonfarm expenditures by 14.6 and 14.3 percent on average, respectively. However, these average effects mask heterogeneity. The farm investment response appears to rise with landholdings: we identify a credit-induced 20.8 percent increase in fertilizer utilization for each additional hectare, a response paralleled imprecisely in total farm expenditures (6.8 percent per hectare, but not statistically significant). In contrast, the effect on nonfarm expenditures does not vary significantly with farm size. This is consistent with previous empirical evidence suggesting that the smallest or liquidity constrained landholders often allocate a disproportionate share of additional liquidity toward nonfarm expenditures rather than farm investment (Hussain and Thapa, 2016; Fink et al., 2020).

These findings suggest that agricultural credit operates through different channels depending on farm scale. For smaller landholders, credit primarily relaxes household liquidity constraints, while for relatively larger smallholders it supports productive intensification. From a policy perspective, the results underscore the limits of uniform credit expansions and highlight the importance of scale-sensitive credit design and complementary risk-mitigation mechanisms.

Our paper contributes to the literature on agricultural credit and smallholder behavior in three ways. First, we provide new evidence on the allocation of agricultural credit across uses in Nigeria by documenting how loan access affects both farm and nonfarm expenditures. Rather than focusing solely on production outcomes, our expenditure-based approach captures multiple channels through which credit relaxes household liquidity constraints, including routine consumption support and farm investment. Second, we contribute to the literature on heterogeneous treatment effects by estimating how expenditure responses vary continuously with farm size, rather than grouping farms into broad size categories. We document evidence consistent with a systematic relationship in which smaller landholders allocate a larger share of liquidity toward nonfarm expenditures, while farmers with greater land endowments increasingly channel credit toward productive farm investment, helping reconcile mixed evidence on the investment impacts of agricultural credit. Third, benchmarking panel-adjusted Double Machine Learning against analogous two-way fixed-effects specifications, we show that the value of nonparametric estimation is context dependent: the two estimators converge on nonfarm routine expenditures but diverge on the farm investment margin, where the household response is nonlinear in scale. This analysis demonstrates how flexible estimation can aid the recovery of policy-relevant, plausibly causal estimates in environments characterized by high-dimensional covariates and non-random program participation. In contexts such as Nigeria, where experimental variation is limited and credit pro-

grams are implemented at scale, our approach offers a practical framework for evaluating food system interventions while accounting for household heterogeneity.

The remainder of the paper is organized as follows. Section 2 describes the institutional context and data. Section 3 outlines the empirical strategy. Section 4 presents results on average and heterogeneous effects of agricultural credit on expenditure allocation. Section 5 discusses the findings and their limitations, and Section 6 concludes and draws policy implications.

2 Study Setting

2.1 Institutional Context

Although Nigeria has long been characterized as Africa's largest oil exporter, oil revenues have declined as a share of GDP over the past three decades. Over the same period, agriculture has become increasingly central to the domestic economy, surpassing oil in its contribution to GDP and accounting for roughly one-quarter of total employment. Despite this economic importance, the agricultural sector remains highly fragmented and dominated by smallholders, with an average farm size of approximately 0.5 hectares. Development challenges are substantial. Survey evidence suggests that nearly 70 percent of smallholder farmers live below the poverty line, most operate at subsistence levels where a large share of output is consumed rather than marketed, and only about 7 percent report access to formal credit (FAO, 2018). In theory, relaxing liquidity constraints in such settings should generate high marginal returns to agricultural investment (Ali et al., 2014; Carter and Olinto, 2003). In practice, however, binding consumption needs and income volatility may lead households to allocate additional liquidity toward food, essential goods, or risk management rather than exclusively toward farm intensification (Karlan et al., 2014; Suri et al., 2021; Bocher et al., 2017). Low credit penetration therefore presents both a constraint and a policy opportunity, and credit expansion has the potential to improve household welfare while strengthening domestic food production.

Nigeria's farm credit market is segmented across formal, public, and informal institutions. Formal providers include commercial banks, microfinance institutions, and public entities. Informal and semi-formal mechanisms—including family networks, cooperatives, and rotating savings associations (e.g., Ajo or Esusu)—play a critical role in filling gaps left by the formal sector (Adetiloye, 2012). Nonetheless, overall credit access remains limited. Formal lenders face high transaction costs, collateral constraints, and information asymmetries that restrict lending to smallholders, while informal lenders lack sufficient capital to meet broader demand (Awotide et al., 2015; Osabohien et al., 2020; Ogebe et al., 2022; Ukwuaba et al., 2020). This segmentation reflects broader patterns documented in the credit literature: marginal enterprises operating at very small scales often prioritize immediate consumption and liquidity needs when credit becomes available,

whereas larger producers are better positioned to translate borrowing into productive investment (Banerjee et al., 2015; Hussain and Thapa, 2016).

In response to these constraints, the Federal Government and the Central Bank of Nigeria have repeatedly implemented directed lending and risk-sharing programs. The Agricultural Credit Guarantee Scheme Fund (ACGSF), established in 1977, guarantees up to 75 percent of loan defaults for participating lenders. Subsequent reforms introduced mechanisms such as the Self-Help Group Linkage Banking model, the Trust Fund Model, and the Interest Drawback Scheme, all intended to reduce effective borrowing costs and encourage lending to smallholders. Other initiatives, including the Family Economic Advancement Programme, sought to expand and partially formalize traditional savings mechanisms (FAO, 2013; Obinne, 1999). Despite these efforts, credit to agriculture remained limited historically, averaging roughly 3 percent of total national credit between 1978 and 2006 (Adetiloye, 2012). More recent evidence for the broader population also indicates limited formal credit use. In a 2024 survey, 9.1 percent of Nigerian adults aged 15 and older reported formal borrowing in the preceding 12 months, while 78.9 percent reported borrowing from formal or informal sources under a broad definition that includes purchasing household food on credit (World Bank Group, 2025b).

More recently, the Anchor Borrowers' Programme (ABP), launched in 2015, aimed to address coordination failures through an interlinked transaction model connecting banks, farmers, and off-takers. The program provided credit guarantees while requiring farmers to sell output to designated buyers to reduce default risk. At its peak, the ABP facilitated over one trillion Nigerian naira (approximately USD 2.7 billion, c. 2020) in loan commitments and has been associated with increases in domestic grain production (Umeh and Adejo, 2019). However, repayment rates were reported to be approximately 25 percent (IMF, 2023), and amid tightening fiscal conditions, new loan applications were suspended in 2024.

These developments occurred alongside substantial macroeconomic shocks. Border closures in 2019 contributed to rising food prices, followed by oil price volatility in 2020 that constrained government revenues. Subsequent fiscal adjustments included the unwinding of programs such as the ABP, the removal of fuel subsidies, and exchange rate reforms. Together, these shocks have intensified food insecurity[1] and heightened pressure on rural households. In this environment, understanding how credit is allocated once accessed is particularly salient. Distinguishing whether loans support agricultural intensification or primarily facilitate routine consumption is central to evaluating the effectiveness of credit-based food system interventions.

[1]For the interested reader, we present evidence of the rising food insecurity among Nigerian households over the period 2018-2024 in Figure A1 of the supplementary Appendix A.

2.2 Living Standards Measurement Study Core Data

We use data from the Nigeria General Household Survey – Panel (GHS-Panel), part of the World Bank's Living Standards Measurement Study–Integrated Surveys on Agriculture (LSMS-ISA). The GHS-Panel is nationally representative and designed for policy analysis. Our analysis draws on the two most recent waves (Wave 4: 2018–2019; Wave 5: 2023–2024), each covering approximately 5,000 households, with roughly 89 percent successfully re-interviewed across waves. Each wave includes household, community, and agricultural questionnaires collected during post-planting (July–September) and post-harvest (January–March) phases (NBS, 2019, 2024).

The household module records demographic characteristics, asset ownership, income sources, economic shocks, and food consumption. We aggregate individual-level information to the household level (the unit of analysis), using head-of-household characteristics as demographic controls. The agricultural module is recorded at the plot level and includes land tenure, plot size, crop production, and fertilizer use. Plot-level variables are aggregated to the household level for key measures such as total cultivated area and fertilizer use.

We construct two primary outcome variables: farm expenditures and nonfarm expenditures. Farm expenditures include land (rent or purchase), livestock (purchase and upkeep), crop inputs (seeds, fertilizer, machinery, pesticides), and hired labor. Nonfarm expenditures include imputed food consumption and non-food expenditures (durables, non-durables, and services), excluding nonfarm business investments, education, health, vehicles, ceremonies, debt, rent, and holidays to isolate regular household consumption. All expenditure variables are annualized to ensure comparability across recall periods.[2]

Key control variables include farm land area, value of crop production, durable assets, livestock holdings, nonfarm income, land rights, economic shock indicators, food security measures, and demographic characteristics.

To construct the analytical sample, we harmonize variable definitions across waves and standardize categorical variables where classifications differ. We restrict our analysis to households engaged in agriculture (2,410 observations dropped by design) and that can be paired across waves (883 observations dropped by attrition), cumulatively reducing the sample from 9,691 to 6,398 observations, corresponding to 3,199 agricultural households observed in both waves. For specifications involving farm-size interactions, we restrict to 5,900 observations with non-missing land area. As discussed in Section 3, nuisance functions are trained using the full paired panel with imputation to maximize predictive performance, while final treatment effect estimates are reported on the restricted subsample to maintain consistency across models. Extreme values in assets, expenditures, and land area are winsorized at the 99th percentile.

Table A2 defines the key variables used in the analysis, Table 1 reports summary statistics for

[2]Food expenditures are based on 7-day recall, while some agricultural expenditures use longer recall windows. All are converted to annual values.

the full sample by wave, and Tables A3, A4, and A5 report quartile summary statistics (pooled and by survey wave). The proportion of landholders receiving agricultural loans is uniformly low across survey waves, with approximately 4.9 and 4.8 percent of households in wave 4 and 5, respectively (full sample N=6,398). Nominal loan values increased between waves 4 and 5, with the mean loan size increasing from ₦76.7 to ₦129.8 thousand. Notably, the second pooled quartile (Table A3) reports the highest mean loan size (₦110.2 thousand), indicating that loan amounts do not increase monotonically with farm size. This is noteworthy given the relatively large differences in landholdings, with the average farm size rising from 0.11 hectares in the bottom quartile to 3.16 hectares in the top quartile (an increase of more than 20-fold across quartiles).[3]

Expenditure patterns differ substantially across the farm size distribution. Households in the top quartile have considerably higher farm expenditures but lower nonfarm expenditures than those in the bottom quartile. Although expenditure profiles vary sharply with land endowments, access to credit remains limited across the distribution, consistent with broad credit rationing.

2.3 Loans, Economic Shocks, and Food Security Data

The LSMS household questionnaire conducted during the post-planting visit records loan applications in the twelve months preceding the survey, approvals, sources, and intended use.[4] Pooling data across waves, roughly one-third of loan applications were earmarked for farming, with approximately 76 percent of agricultural loans sourced from informal or semi-formal sources (e.g. friends, money-lenders, savings associations).[5]

To approximate the effect of agricultural credit programs (such as the Anchor Borrowers Program), we use loans specifically earmarked for farm expenditures as a proxy for agricultural credit.[6] We pool loans across informal and formal sources for two reasons: (i) both sources are effectively fungible in relaxing the household liquidity constraint, and (ii) the scarcity of formal agricultural loans necessitates pooling to preserve statistical power.

We define credit access at the extensive margin (a binary treatment variable), because relaxing binding budget constraints often induces structural changes in household expenditure behavior, such as 'crowd-in' effects (Banerjee et al., 2019). Furthermore, this approach mitigates endogeneity concerns regarding the correlation between loan magnitude and household assets. Additionally, to isolate the treatment effect and account for households with multiple loans, we include a control dummy variable for non-agricultural loans. We choose to exclude non-agricultural loans from the

[3] More details about the distribution of farm size by loan access can be found in Figure A3 of the supplementary Appendix A.

[4] The specific wording of the survey questionnaire is "In the last 12 months, have you or anyone else in the household attempted to borrow money or applied for or received a loan from [a list of plausible sources]?" There is no indication that loans were made "in-kind" and specific values in Naira are provided for each loan received.

[5] See Figure A4 in Appendix A for more details on the distribution of farm size by loan formality.

[6] The LSMS datasets do not include information on whether a loan originates from a specific government program, or if it otherwise could be formally defined as an agricultural loan.

TABLE 1.
Summary Statistics by Wave

	Sample Mean			Min	Max
	Wave 4	Wave 5	Pooled		
Agricultural Loan Variables					
Received Agricultural Loan (Binary 0/1)	0.05	0.05	0.05	0.00	1.00
Size of Agricultural Loan (1,000s ₦) (Recipients Only)	76.72	129.83	102.85	2.00	1,500.00
Outcome Variables					
Input Expenditures (1,000s ₦)[a]	26.40	40.01	33.20	0.00	501.50
Land Expenditures (1,000s ₦)[a]	4.57	4.02	4.29	0.00	215.00
Labor Expenditures (1,000s ₦)[a]	66.87	62.27	64.56	0.00	1,052.25
Livestock Expenditures (1,000s ₦)[a]	19.61	47.79	32.31	0.00	796.00
Farm Expenditures (1,000s ₦)	106.99	128.98	117.98	0.00	1,385.15
Food Expenditures (1,000s ₦)[a]	1,338.82	1,737.89	1,538.36	0.00	66,861.29
Non-Food Expenditures (1,000s ₦)[a]	287.87	567.46	427.67	3.24	9,163.29
Nonfarm Expenditures (1,000s ₦)	1,626.70	2,305.36	1,966.03	13.12	67,818.62
All Fertilizer Use (kg/ha)	872.35	633.75	753.05	0.00	47,237.92
Ratios					
Agricultural Loan to Total Farm Expenditures	71.7%	100.7%	87.2%		
Agricultural Loan to Nonfarm Expenditures	4.7%	5.6%	5.2%		
Control Variables					
Farming Land Area (Hectares)[a,b]	1.40	0.97	1.18	0.01	8.00
Value of Farm Production (1,000s ₦)[a,b]	338.65	372.75	355.75	0.00	3,606.00
Assets (1,000s ₦)[a,b]	193.05	360.31	276.68	0.00	7,735.00
Nonfarm Income (1,000s ₦)[a,b]	272.69	427.05	349.87	0.00	5,520.00
Livestock Holdings (#)[a,b]	0.93	0.65	0.79	0.00	11.90
Has Formal Land Rights (binary 0/1)[a,b]	0.17	0.12	0.14	0.00	1.00
Income Shock (#)	0.25	0.15	0.20	0.00	5.00
Production Shock (#)	0.38	1.52	0.95	0.00	8.00
Price Shock (#)	0.19	1.94	1.07	0.00	5.00
Head Marital Status (binary 0/1)[c]	0.78	0.76	0.77	0.00	1.00
Head Age[c]	50.02	53.47	51.74	10.00	130.00
Head Sex (binary 1 ≡ Male)[c]	0.83	0.79	0.81	0.00	1.00
Number of Members in the Household	5.91	5.94	5.93	1.00	29.00
Number of Adult Members in the Household	2.88	2.98	2.93	0.00	16.00
Household Has Phone Access (binary 0/1)	0.94	0.92	0.93	0.00	1.00
Household Has Internet Access (binary 0/1)	0.36	0.37	0.36	0.00	1.00
Probability of Moderate Food Insecurity	0.33	0.58	0.46	0.03	1.00
Food Consumption Score	51.43	46.61	49.02	0.00	112.00
Received Non-Agricultural Loan (binary 0/1)	0.10	0.07	0.09	0.00	1.00

[a] Winsorized to top 1%.
[b] Imputation performed by Evans School Policy Analysis and Research Group (UW EPAR, 2025).
[c] Characteristics of the Head of Household, coded verbatim from survey responses.

treatment variable to more closely proxy the effects of formal agricultural loans rather than general household liquidity relaxation.

To control for exogenous disturbances, we use the Economic Shocks module within the household questionnaire during the post-harvest visit, which records unexpected events affecting the

household's welfare in the three years preceding the survey. Because the specific shock categories vary slightly between waves (21 categories in Wave 4 and 28 in Wave 5), we harmonize them into four distinct categories: (1) Production Shocks (e.g. drought, floods, pests), (2) Income Shocks (e.g. death of earner, loss of remittances), (3) Price Shocks (e.g. fuel price inflation, input/output price volatility), (4) Other Shocks (e.g. conflict, theft).[7] We restrict the shock window to the two years preceding the survey wave to reflect acute events which might induce or be ameliorated by credit acquisition. Specifically, we sum the number of shocks experienced by the household in each category to capture the acute intensity of shocks experienced by the household not absorbed by fixed effects.

Finally, we control for food security status. We measure household food security using the Food Consumption Score and the Food Insecurity Experience Scale (FIES), calculated using data from the Food Security module present in the household questionnaire post-harvest visit in both waves. For FIES, we follow Cafiero et al. (2018) and estimate the probability of a household experiencing either moderate or severe food insecurity.

3 Empirical Framework

3.1 Estimation Strategy

Our objective is to estimate how access to agricultural credit affects household expenditure allocation, and whether these effects vary with farm size. We exploit the panel structure of the LSMS data, using within-household variation between Wave 4 (2018–2019) and Wave 5 (2023–2024) to control for time-invariant unobserved heterogeneity.

A conventional two-way fixed effects (TWFE) model would address household-level fixed characteristics and common time shocks. However, given the high dimensionality of observable covariates and the potentially complex selection into credit, we adopt a Double/Debiased Machine Learning (DML) framework[8]. DML operates by partialling out the confounding effects of observables from both outcome and treatment variables, leveraging high-dimensional machine learning functions to isolate pseudo-random variation in credit access[9]. This approach preserves the fixed-effects structure while allowing flexible, non-parametric modeling of the relationships between covariates, treatment assignment, and outcomes. In our results section, and in supplementary tables in Appendix A, we benchmark our panel-adjusted DML against analogous TWFE specifications.

[7] The Other Shocks category is not included in the analysis due to its extremely sparse prevalence and no detectable effect on nuisance function performance.

[8] For our analysis, we primarily utilize the DoubleML package developed for Double Machine Learning in Python (Bach et al., 2022).

[9] DML necessarily relies upon the assumption of conditional independence; as such, we explore the robustness of our results to omitted variable bias in Section 4.4.2.

In the following subsections, we outline the DML specification, its adaptation to panel data, and the implementation details of our estimation procedure.

3.2 Identification

3.2.1 TWFE Benchmark and DML Specification

A conventional econometric identification method in this context leverages the panel data to control for household- and time-level heterogeneity with a TWFE specification. Specifically, the TWFE specification demeans the sample with linear household and time fixed effects to purge unobserved variable bias. Additionally, TWFE incorporates linear controls to improve precision and—depending on the analysis—assert conditional independence. Throughout our Results Section we utilize a TWFE specification as a linear benchmark to our DML results, defined as:

$$Y_{it} = \alpha_i + \lambda_t + \theta_1 L_{it} + \theta_2 \left(L_{it} \times \hat{A}_{it}\right) + \gamma X_{it} + u_{it} \tag{1}$$

where Y_{it} is the outcome and denotes each category of expenditure for household i at time t (Expenditures), α_i are household fixed effects, λ_t are the survey wave fixed effects, and L_{it} is the treatment and denotes whether or not the household received an agricultural loan (Received Agricultural Loan). $\hat{A}_{it}$ represents the centered farm size in hectares (Farming Land Area) defined as $\hat{A}_{it} = A_{it} - \bar{A}$, where $\bar{A}$ is the sample mean across all households i and survey waves t. Finally, X_{it} is the matrix of time-varying control variables capturing observable confounding for each category of expenditure and u_{it} is the error term.

In this paper, we assert that smallholder expenditure decisions are highly complex and often nonlinear in relation to observables. To that end, we isolate the causal effect of agricultural loan use on household expenditures while accounting for the high-dimensional confounders pervasive in credit acquisition, by employing a Double/Debiased Machine Learning (DML) approach. Unlike the linear TWFE benchmark, DML allows for non-parametric modeling of the relationship between covariates, treatment assignment, and outcomes, which can improve identification when compared to traditional econometric techniques (Belloni et al., 2014; Chernozhukov et al., 2018, 2025; Semenova and Chernozhukov, 2021). This flexibility is useful in our setting given the largely informal and opaque credit markets accessible to smallholders.

We estimate a Partial Linear Regression (PLR) model with an interaction term, defined as:

$$Y_{it} = \theta_1 L_{it} + \theta_2 \left(L_{it} \times \hat{A}_{it}\right) + g\left(X_{it}, \bar{X}_i\right) + u_{it} \tag{2}$$

In equation 2, fixed effects (α_i and λ_t) and linear controls (γX_{it}) are replaced with the unobserved, non-parametric function $g(\cdot)$, known as the "nuisance function". The nuisance function captures confounding variance for each category of expenditure separately through both time-

varying controls X_{it} and their Mundlak means $\bar{X}_i$, calculated for each household over time, controlling for time-invariant heterogeneity—the procedure of nuisance function estimation and incorporation of fixed effects are expanded upon in greater detail in Sections 3.2.2 and 3.2.3, respectively. We first estimate the model without the interaction term to obtain the baseline treatment effect, and then include the interaction term to test for heterogeneity based on farm size. Following Chernozhukov et al. (2025), we treat the interaction term $\left(L_{it} \times \hat{A}_{it}\right)$ as a conditional average treatment effect (CATE) such that $\hat{A}_{it} \in X_{it}$. This allows us to jointly estimate the baseline average treatment effect θ_1, and the heterogeneous treatment effect θ_2. The equations are estimated separately for each category of farm or nonfarm expenditures, and then for each of the sub-categories in farm expenditures (input, land, labor, livestock expenditures) and nonfarm expenditures (food and non-food expenditures).

Each expenditure category comprises annualized expenditures recorded in the LSMS-ISA survey. Input expenditures consist of items such as seed, pesticide, and fertilizer; land expenditures consist of land purchases in the prior year and annualized agricultural plot rents; labor consists of hired human and rented animal labor; and livestock consists of purchases of animals such chickens, cows, or donkeys. In comparison, nonfarm expenditures consist of routine household expenditures not directly related to agricultural activities: food consists of 7-day recall purchased food consumption; non-food consists of a wide variety of routine expenditures such as clothing, fuel, or electronics.

3.2.2 Stacked Learners and Rich Controls

In the DML framework, the conditional expectation of the outcomes $\hat{E}[Y|X]$ and the treatment $\hat{E}[L|X]$ serve as nuisance function parameters labeled $\hat{n}(\cdot)$ and $\hat{m}(\cdot)$ respectively — that is, functions that must be estimated to partial out confounding but are not the core objective of the analysis. Formally, DML is a two-stage estimation process, first nuisance functions are estimated via cross-fitting (described in Section 3.2.3) and partialled out

$$\tilde{Y}_{it} = Y_{it} - \hat{n}(X_{it}, \bar{X}_i) \quad \text{and} \quad \tilde{L}_{it} = L_{it} - \hat{m}(X_{it}, \bar{X}_i). \tag{3}$$

The nonparametric partialling out methodology has its canonical origins in the seminal Robinson (1988), with important later developments including post-double-selection (Belloni et al., 2014) and double/debiased machine learning (Chernozhukov et al., 2018). The purpose of the partialling out step is to isolate variation in treatment that is not predictable on observables, while removing the corresponding variation in outcomes. Additionally, partialling out variation in both treatment and controls prior to treatment effect yields Neyman-orthogonality. Effectively, this means that point estimates produced in the second stage (Equation 4) are robust to minor errors in nuisance function estimation, which would otherwise bias results if the nuisance function was

estimated simultaneously with treatment effects (i.e. $Y_{it} = \theta_1 L_{it} + \theta_2 \left(L_{it} \times \hat{A}_{it}\right) + \hat{g}(X_{it}, \bar{X}_i)$).

$$\tilde{Y}_{it} = \theta_1 \tilde{L}_{it} + \theta_2 \left(\tilde{L}_{it} \times \hat{A}_{it}\right) + u_{it}. \tag{4}$$

Therefore, the performance of the DML estimator hinges on the predictive accuracy of our machine learning models used to approximate these functions. Consequently, to maximize model performance, we implement a specific training and estimation strategy.

The DML estimator is built on distinct estimation samples, which we describe before turning to learner specification. To maintain consistency across model specifications and maximize first-stage accuracy, we estimate $\hat{E}[L|X]$ ($\hat{m}(\cdot)$) once on the full paired panel ($N = 6,398$ pooled observations) and reuse it across outcome-specific models, maximizing the training sample and following the Group ATE framework in Chernozhukov et al. (2025). The outcome functions ($\hat{n}(\cdot)$) are estimated separately for each expenditure category on the paired subsample with non-missing farm size ($N \leq 5,900$ pooled observations), ensuring a common sample across the baseline and heterogeneous specifications.

To use the full information set, we apply median imputation with missingness indicators: each missing control is replaced with the relevant sample median, and a companion dummy flags the imputed entries for each applicable variable. The relevant median depends on the estimation sample: for treatment function estimation (full sample, $N = 6,398$) we impute farm size ($N = 318$), value of crop production ($N = 318$), formal land rights ($N = 815$), and household head marital status ($N = 18$); for the restricted outcome sample we impute formal land rights, and household head marital status with the precise number of imputed observations depending on the outcome sample (e.g. $N = 597$ and $N = 18$ for the baseline nonfarm sample).

When training the nuisance functions we employ a "stacked" ensemble for our learners by training several different types of models and a rich set of covariates in X_{it}. Ensemble learners are nested ML functions composed of a meta-learner and stacked learners, for example

$$\hat{m}(\cdot) = \hat{f}(\hat{r}(\cdot), \hat{t}(\cdot), \hat{w}(\cdot)) \tag{5}$$

where $\hat{f}(\cdot)$ is a meta-learner such as ridge, and $\hat{r}(\cdot)$, $\hat{t}(\cdot)$, $\hat{w}(\cdot)$ are stacked learners such as random forest, gradient boosting, or regularized models.

We use ensemble learners as a best practice technique for nuisance function estimation to improve the stability and generalization of our results. Our ensemble integrates a diverse set of algorithms to balance bias and variance: (i) Tree Based Models, including Random Forest and Histogram Gradient Boosting to capture non-parametric relationships, and (ii) Regularized Linear Models, including Lasso, Ridge, and Elastic Net to handle high-dimensionality and overfitting.

We use distinct learners for our outcome and treatment variables. For the binary agricultural loan treatment, we use classification variants of the learners (e.g. Logistic Regression with Elas-

tic Net penalty, and stacking classifier), while we employ their regression counterparts for the continuous expenditures outcome. The meta-learners for each ensemble are a Ridge model for the continuous outcome and a Logit for the binary treatment. Finally, to compute the interaction term, the residual agricultural loan treatment variable is interacted with the centered farm size ($\hat{A}$).

Complementing this robust learner set, X_{it} comprises a rich vector of socio-economic and fixed effect variables. As we mentioned above, to account for time-invariant household heterogeneity, we calculate Mundlak means for all time-varying controls and cluster standard errors at the household level. Finally, we include state and time fixed effects, corresponding to the Nigerian Federal states and the wave indicator variables, respectively.

3.2.3 DML in Panel Data

Adapting DML to a panel dataset requires careful handling of unobserved heterogeneity. Traditional DML assumes independence of observations, which is violated by repeated household observations as is the case here. Additionally, recovering within-household estimates—as in TWFE—from DML is challenging as demeaning data prior to estimation can introduce bias. To address these challenges, we implement two modifications aligned with the best practices from the emerging panel DML literature.

First, we incorporate Correlated Random Effects (CRE) by augmenting the control matrix X_{it} with household-level time means (Mundlak means):

$$\bar{X}_i = \frac{x_{i1} + x_{i2}}{2}. \tag{6}$$

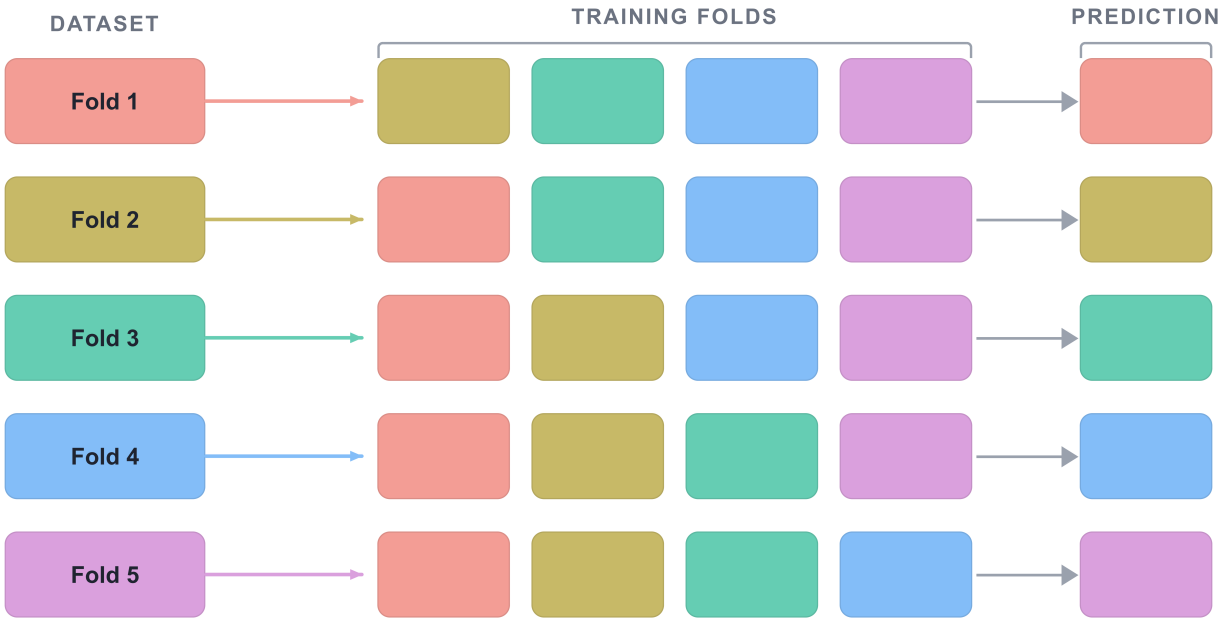
We utilize Mundlak means to approximate the within-household treatment effect estimates produced by a TWFE specification. Specifically, when estimating treatment effects in a TWFE specification, household level fixed-effects are demeaned—eliminating α_i in Equation 1—before estimating treatment effects (i.e. $\tilde{y}_{it} = y_{it} - \frac{1}{T}\sum_{t=1}^{T} y_{it}$ and $\tilde{x}_{it} = x_{it} - \frac{1}{T}\sum_{t=1}^{T} x_{it}$). This procedure controls for time-invariant heterogeneity at the household level, yielding within-household treatment effect estimates.

Although the demeaning procedure is widely utilized for analysis of observational panel data, if done prior to nuisance function estimation it can yield biased results in DML (Fuhr and Papies, 2024). Thus, to control for household level time-invariant heterogeneity, incorporating Mundlak means for time varying covariates into the control set has emerged as a common practice (Fuhr and Papies, 2024; Clarke and Polselli, 2026). The nuisance function is therefore written as $g(X_{it}, \bar{X}_i)$, where $\bar{X}_i$ denotes the vector of household-level means.

Similarly, a TWFE specification controls for inter-temporal heterogeneity with time fixed-effects (represented by λ_t in Equation 1). Following a similar reasoning to the inclusion of Mundlak means in the control set (X_{it}), it is standard practice to include time dummies in X_{it}. Therefore, we

include a dummy for the second time period (second wave of the survey) inside X_{it} to account for period specific global shocks. Taken together, these modifications allow DML to produce results analogous to a traditional TWFE model, but with the non-parametric flexibility of DML.

Second, we enforce grouped cross-fitting. Cross-fitting, illustrated in Figure 1, is a standard technique for avoiding overfitting. The dataset is first divided into subsets ("folds"); a model is trained on one group of folds and predictions are made on the held-out fold. For example, in Figure 1, row 1, a supervised machine learning (ML) model is trained on Folds 2 through 5 (k_2, k_3, k_4, k_5) and used to predict Fold 1 (k_1). The goal is to avoid the bias created when ML models fit idiosyncratic noise in the training data, referred to as overfitting.



Notes: In each row, nuisance models are trained on four folds and used to predict the held-out fold for both the outcome variable and treatment propensity. Folds are grouped at the household level, with both waves of a household assigned to the same fold.

FIGURE 1

To illustrate, suppose for simplicity we have a cross-sectional sample with 1,000 observations, including an outcome Y_i, a treatment L_i, and controls X_i. In a five-fold cross-fitting specification, the sample is divided equally into five mutually exclusive folds ("subsets") of 200 observations each. Nuisance function estimation and partialling out (explained in greater detail in Section 3.2.2) proceeds in five stages. In the first stage (row 1 of Figure 1), fold k_1 is held-out (i.e. $-k_1$) and we estimate the nuisance functions with supervised machine learning ($\hat{m}_{-k_1}(\cdot) = \hat{E}_{-k_1}[L_{-k_1}|X_{-k_1}]$ and $\hat{n}_{-k_1}(\cdot) = \hat{E}_{-k_1}[Y_{-k_1}|X_{-k_1}]$) utilizing folds 2 through 5 (i.e. the "training folds"). We then estimate the outcome and treatment predictions for fold 1 using the nuisance functions estimated on the training folds ($\hat{L}_{k_1} = \hat{m}_{-k_1}(X_{k_1})$ and $\hat{Y}_{k_1} = \hat{n}_{-k_1}(X_{k_1})$). This procedure is then repeated over the

four remaining folds, holding out the fold of interest from the training folds, and taking the union of each stage ($\hat{L} = \bigcup_{l=1}^{5} \hat{L}_{k_l}$ and $\hat{Y} = \bigcup_{l=1}^{5} \hat{Y}_{k_l}$) to generate the final nuisance function prediction set with 1,000 observations.

By predicting outcome and treatment values in fold k_l with a nuisance function trained on a mutually exclusive set of observations ($\hat{m}_{-k_l}(\cdot)$ and $\hat{n}_{-k_l}(\cdot)$), cross-fitting avoids overfitting bias produced when machine learning algorithms fit idiosyncratic noise to outcomes. However, the cross-fitting procedure relies on the independence of observations when dividing data into folds to ensure idiosyncratic noise is uncorrelated between folds. Specifically, independence between folds implies overfitting bias from the training folds ($-k_l$) is orthogonal to the prediction folds (k_l). To address within-household inter-temporal correlation in our panel data we group data at the household level when creating folds (Clarke and Polselli, 2026). Formally, given a household with observations i_1 and i_2 in waves 1 and 2 respectively, we require

$$i_1 \in k_l \iff i_2 \in k_l \tag{7}$$

for each fold k_l. So, both observations for the same household are always assigned to the same fold. This grouped cross-fitting procedure maintains independence across folds during nuisance function estimation.

3.2.4 Inference and Robustness

To ensure the stability of our results, we repeat the cross-fitting procedure over 30 randomized repetitions and aggregate results using the median (Chernozhukov et al., 2018, 2025). We assess predictive validity of the nuisance functions using three primary diagnostics. For the outcome model, we report a standard R^2 (R^2_Y) value. For the treatment propensity model (i.e. the likelihood of getting a loan, expressed by $\hat{m}(\cdot)$), we report the Area Under the Curve (AUC) and Average Precision (AP). We select these complementary metrics to address the specific challenges of our data. AUC[10] provides a standard measure of discrimination often used in traditional econometrics; a naive value of 0.5 indicates the model is no better than chance at ranking a treated observation above an untreated one. By contrast, AP[11] offers a more informative benchmark under treatment imbalance—as in our context—since its naive value equals the sample prevalence of treatment, in this case 4.9% (0.049). Values above these naive baselines indicate the learners are identifying gen-

[10] The Area Under the Curve measures the probability a model correctly ranks a randomly chosen treated observation higher than a randomly chosen untreated observation. A value of 0.5 indicates predictive power equivalent to random choice, while a value greater than 0.5 indicates the model assigns treatment probabilities more accurately than random chance.

[11] Average Precision summarizes a model's precision recall curve over all decision thresholds ($p(treatment|X) \geq \tau,\ \tau \in [0,1]$). A naive model that assigns treatment at random will yield an AP equal to the sample prevalence of treatment. Therefore, AP is beneficial to contextualize the model's predictive power in settings such as ours where treatment is unbalanced and rare.

uine signals rather than noise. Finally, we perform two distinct robustness checks: first, to ensure our estimates are robust to fold specification, we test the robustness of our baseline 5-fold estimates against alternative 3- and 8-fold specifications in Section 4.4.1; second we perform a pseudo omitted variable analysis in Section 4.4.2.

3.3 Log Transformation and Intensive Margin Interpretation

Our primary estimand of interest is the average household level response in log expenditures rather than the average change in expenditure levels. Household expenditures are strongly right skewed, with a heavy tailed distribution as shown in Figure A2. Consequently, an ATE in levels ($\mathbf{E}[Y(1) - Y(0)]$) can be disproportionately influenced by a small number of households with large expenditures. We use log transformation outcomes, which reduce the influence of households with large expenditures and more closely align the estimand with household-level proportional changes. This choice reflects our interest in average proportional treatment responses across households while reducing the influence of high spending households on the estimand. That said, log transformation introduces important limitations for estimation and interpretation, and we follow the guidance of Chen and Roth (2024) in addressing these limitations.

Most notably, the standard log transformation requires restricting the analysis to observations with positive expenditures. Households reporting no expenditures in a given category are therefore excluded, resulting in a modest sample reduction for the headline aggregate expenditure analysis (Tables 2 and 3), and a more substantial reduction in sample size for our disaggregated farm expenditure analysis (Table 4).[12]

A common alternative is to retain zero valued observations with log-like transformations such as $\log(Y + 1)$ or $\text{arcsinh}(Y)$. However, Chen and Roth (2024) show that, when treatment affects positive outcomes, treatment effects estimated using these transformations are sensitive to the units in which the outcome is measured. Effectively, the scale of the outcome variable determines the weight assigned to the extensive margin when estimating treatment effects. We therefore prefer the standard log transformation, accepting the reduction in sample size, rather than adopting a log-like transformation whose scaling implicitly determines the relative weight of extensive and intensive margin responses.

Restricting the sample to observed positive expenditures yields an estimand that is closely related to, but not generally identified as, a pure intensive margin effect (Chen and Roth, 2024). That

[12]Specifically (after removing observations for missing farm size values), among the aggregate expenditure variables: zero observations are dropped due to nonfarm expenditure zero valued observations, and 212 households ($N = 424$) are dropped due to farm expenditure missingness in the baseline and heterogeneous analysis. In the disaggregated expenditure analysis: Zero households and one household ($N = 2$) are dropped because of nonfarm non-food and food expenditure zeros, respectively; 1,115 ($N = 2,230$), 2,774 ($N = 5,548$), 945 ($N = 1,890$), and 2,362 ($N = 4,724$) households are dropped due to farm input, land, labor, and animal expenditure missingness respectively. Finally, 2,112 ($N = 4,224$) households are dropped due to fertilizer utilization missingness.

is, among households with positive expenditures, we cannot distinguish households that would have positive expenditures regardless of treatment from those who entered on the extensive margin. This could include, for example, a household that chose to make farm expenditures only after receiving a loan. To reflect this, in our analysis we utilize the terminology of *quasi-intensive-margin effects*: treatment effects on log expenditures within the observed positive outcome sample that may also reflect treatment induced changes in the composition of that sample (extensive margin effects).

The potential interpretive concern is plausibly less consequential for aggregate outcomes (Tables 2 and 3), for which zero expenditures are uncommon (0.0% for nonfarm expenditures and 7.2% for farm expenditures). In contrast, this limitation is plausibly more consequential for our disaggregated farm expenditure analysis (Table 4), where the prevalence of zero expenditures is much greater (e.g. 94.0% for farm land expenditures and 32.0% for farm labor expenditures).

4 Results

This section presents the empirical results from the DML models benchmarked against analogous TWFE specifications. We interpret the estimates as reflecting changes in household expenditure allocation after receiving agricultural credit. Because credit relaxes liquidity constraints at the household level, observed expenditure responses capture reallocation across farm and nonfarm uses rather than adherence to stated loan purposes. We first report baseline average treatment effects, then examine heterogeneity by farm size, followed by disaggregated expenditure analyses and robustness checks.[13] Note that calculations in this section may deviate from direct analysis using table output due to rounding.

4.1 Baseline results

Table 2 reports the baseline average treatment effects (ATE) of receiving an agricultural loan on household expenditures. Because the outcomes are log-transformed and treatment is binary, coefficients are interpreted as percentage changes using $100 \times (e^{\beta} - 1)$. Because observations with zero expenditures are excluded, estimates reflect *quasi-intensive-margin* responses for households with positive expenditures (as described in Section 3.3).

Our DML results indicate that receiving an agricultural loan increases farm and nonfarm expenditures by 14.6 and 14.3 percent, statistically significant at the 10 percent and 1 percent levels, respectively. These estimates exploit within-household variation across survey waves and control for observed covariates, household correlated random effects, and common time shocks.

These results are consistent with experimental evidence showing that relaxing an agricultural

[13] A balance table and additional descriptive figures are reported in Appendix A Table A1.

TABLE 2.
Average Treatment Effects of Agricultural Loans on Expenditures

	Farm Expenditures (log)		Nonfarm Expenditures (log)	
	DML	TWFE	DML	TWFE
Received Agricultural Loan	0.137*	0.073	0.134***	0.154**
	(0.071)	(0.095)	(0.049)	(0.070)
Specification				
Controls	Yes	Yes	Yes	Yes
Federal State FEs	Yes	Yes	Yes	Yes
Survey Wave FEs	Yes	Yes	Yes	Yes
Household CREs	Yes	No	Yes	No
Household FEs	No	Yes	No	Yes
Learners				
Loan $\sim X$	PLH-L		PLH-L	
Expenditures $\sim X$	ERH-R		ERH-R	
K-folds	5		5	
Diagnostics				
R^2_Y (Outcome: $Y \sim X$)	0.351		0.420	
AUC (Loan $\sim X$)	0.621		0.625	
AP (Loan $\sim X$)	0.081		0.079	
Adjusted-R^2		0.352		0.429
Within Adjusted-R^2		0.074		0.089
Observations	5,476	5,476	5,900	5,900

Standard errors in parentheses. Columns 1 and 3 are Double Machine Learning (PLR), $n_{\text{rep}} = 30$; columns 2 and 4 are TWFE with cluster-robust standard errors at the household level.
Adj-R^2 and Within Adj-R^2 report TWFE model fit.
R^2_Y reports nuisance learner performance for the outcome model.
AUC (Area Under Curve) and AP (Average Precision) report nuisance learner performance for the treatment model.
ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.
PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked with a Logistic meta-learner.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

household's budget constraint raises both agricultural investment and nonfarm spending. For example, Delavallade and Godlonton (2023) show that warrantage (inventory credit) in Burkina Faso increases total per-capita expenditures by 13 percent (driven by non-food categories) and raises crop revenues by roughly 15 percent. Similarly, Beaman et al. (2023) find sizable expenditure responses to liquidity provision in Mali.

By comparison, the TWFE estimates indicate no statistically significant effect on farm expenditures and a 16.8 percent increase in nonfarm expenditures, statistically significant at the 5 percent level. These results are informative as a linear benchmark for our primary DML specifica-

tion. For nonfarm expenditures, the TWFE and DML point estimates are closely aligned, suggesting that the positive consumption response is not heavily driven by the nonparametric relationships captured by DML. This is similarly consistent with the expectation that routine household consumption—recall nonfarm expenditures represent daily consumption excluding lumpy consumption (Table A2)—is more mechanically related to observable household needs and liquidity constraints. That is, we posit nonfarm expenditures are driven by observable characteristics (e.g. household size, food requirements, and cash on hand) in a way a linear specification can capture, not a literal passthrough of loan proceeds (which we discuss in Section 5).

By contrast, farm expenditure responses may depend on a more complex and interdependent set of factors, such as farm scale, land quality, or prior success. These relationships are difficult to model parametrically and may be nonlinear or interactive in a way that a TWFE specification cannot fully capture. The difference between the TWFE and DML estimates for farm expenditures therefore suggests that the flexible nuisance function partialling out procedure in DML is most valuable for the farm expenditure margin, where the expenditure response is likely to be less mechanical. We interpret the close alignment on nonfarm as indicating that the nonparametric specification has limited additional value where the response is mechanical, but explore its potential value on the farm margin in the heterogeneity and disaggregation results below where the relevant evidence lies.

This characterization is also supported by the relative magnitudes of variance captured in outcomes by the nuisance function (R^2_Y) and the TWFE estimates (Adjusted-R^2)[14]. Both the DML and TWFE specifications capture significantly more variance in nonfarm expenditures than farm expenditures (0.420 vs 0.351 and 0.429 vs 0.352, respectively). These statistics highlight three important insights: first, our adjustments to DML for panel data effectively capture comparable levels of variance to a TWFE specification when substituting CREs for household fixed-effects; second, our learner ensemble and a linear fixed-effects model closely agree that nonfarm expenditures can be more effectively predicted than farm expenditures with the available controls; and third, despite controlling for comparable variance in the partialling out and prediction steps for both outcomes, point estimates only differ materially for farm expenditure coefficients. This further refines our speculation that nonfarm routine expenditures see limited benefit from nuisance function flexibility, while nonparametric nuisance function estimation may be particularly advantageous for farm expenditures.

The predictive validity of our nuisance functions is substantiated by the Area Under the Curve (AUC) and Average Precision (AP) statistics reported in Table 2. The AUC is a common diagnostic statistic for binary outcome models, and can generally be interpreted as the probability that a

[14]Note R^2_Y and Adjusted-R^2 serve similar but distinct purposes. The R^2_Y statistic characterizes nuisance function fit, not incorporating the treatment effect, while Adjusted-R^2 captures the overall TWFE specification fit, incorporating the treatment effect.

random positive instance is ranked higher than a negative one. Our AUC values of 0.621 and 0.625 (farm and nonfarm expenditure models, respectively) outperform the naive (random) assignment model benchmark of 0.5. While these scores suggest relatively poor discrimination by traditional standards (Hosmer et al., 2013), they align with recent literature on the difficulty of predicting credit behavior in opaque and noisy smallholder microdata. For instance, Björkegren and Grissen (2020) demonstrate that machine learning models constrained to traditional demographic data often yield AUC scores between 0.51 and 0.57 for credit repayment outcomes.

The AP is a standard machine learning metric that summarizes the precision-recall curve. It can generally be interpreted as the average probability that a positive prediction made by the model is correct, benchmarked against the baseline probability of random selection (sample prevalence). Our AP values of 0.081 and 0.079 (farm and nonfarm models, respectively) indicate moderate discrimination above random assignment. Given the low prevalence of agricultural credit in our sample (4.9 percent), a naive model would achieve an AP of only 0.049. Thus our models achieve a precision roughly 62.5–65.6 percent higher than a naive benchmark. Collectively, these diagnostics indicate that despite the opaque nature of informal credit markets where soft variables (e.g. trust, reputation) heavily influence credit access, our learners successfully isolate systemic selection signals necessary for DML nuisance function estimation.

4.2 Heterogeneous treatment effects by farm size

We next examine whether the impact of agricultural credit varies with farm size. Table 3 augments the baseline specification with an interaction between loan receipt and farm land area (centered at the sample mean), providing suggestive evidence that farm expenditure responses increase with farm size. To supplement our analysis, we also conduct a DML quartile-based group average treatment effect (GATE) analysis, shown in Appendix B.

The baseline coefficient on loan receipt remains positive and statistically significant for farm expenditures (0.125 log points, approximately 13.4 percent). The coefficient on the interaction term is also positive (0.066) but not statistically significant at traditional levels (P-value=0.145), suggesting that the farm expenditure response increases by 6.8 percentage points for each additional hectare of land, although it lacks the statistical precision to be distinguished from zero. In contrast, the interaction term for nonfarm expenditures is negative, notably smaller in magnitude (–0.031), and statistically insignificant (P-value=0.314), suggesting that the estimated 14.9 percent increase (0.139 log points) in nonfarm expenditures does not vary systematically with farm size.

The TWFE estimates provide a useful linear benchmark for these heterogeneous effects. Mirroring the baseline estimates in Table 2, the nonfarm expenditure DML and TWFE estimates are closely aligned: both specifications indicate a large significant ATE (0.139 and 0.159, respectively) and a small negative heterogeneous treatment effect (HTE) (-0.031 and -0.029, respectively). This

TABLE 3.
Heterogeneous Effects of Agricultural Loans on Expenditures

	Farm Expenditures (log)		Nonfarm Expenditures (log)	
	DML	TWFE	DML	TWFE
Received Agricultural Loan (Loan)	0.125*	0.045	0.139***	0.159**
	(0.072)	(0.097)	(0.049)	(0.070)
Loan × Farming Land Area (HA)	0.066	0.142**	−0.031	−0.029
	(0.045)	(0.066)	(0.031)	(0.041)
Specification				
Controls	Yes	Yes	Yes	Yes
Federal State FEs	Yes	Yes	Yes	Yes
Survey Wave FEs	Yes	Yes	Yes	Yes
Household CREs	Yes	No	Yes	No
Household FEs	No	Yes	No	Yes
Learners				
Loan $\sim X$	PLH-L		PLH-L	
Expenditures $\sim X$	ERH-R		ERH-R	
K-folds	5		5	
Diagnostics				
R^2_Y (Outcome: $Y \sim X$)	0.351		0.420	
AUC (Loan $\sim X$)	0.621		0.625	
AP (Loan $\sim X$)	0.081		0.079	
Adjusted-R^2		0.353		0.428
Within Adjusted-R^2		0.075		0.089
Observations	5,476	5,476	5,900	5,900

Standard errors in parentheses. Columns 1 and 3 are Double Machine Learning (PLR), $n_{\text{rep}} = 30$;
columns 2 and 4 are TWFE with cluster-robust standard errors at the household level.
Adj-R^2 and Within Adj-R^2 report TWFE model fit.
R^2_Y reports nuisance learner performance for the outcome model.
AUC (Area Under Curve) and AP (Average Precision) report nuisance learner performance for the treatment model.
ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.
PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked with a Logistic meta-learner.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

supports the interpretation that agricultural credit relaxes household liquidity constraints for routine expenditures in a way that is broadly similar across the farm size distribution.

For farm expenditures the comparison is more nuanced, and estimators diverge in both level and slope: the ATE is positive and significant under DML (0.125, P-value=0.082) but small and insignificant under TWFE (0.045), while the linear interaction is significant under TWFE (0.142, P-value=0.032) but imprecise under DML (0.066, P-value=0.145). Having benchmarked the baseline DML estimates against TWFE, we now consider the source of their divergence on the farm mar-

gin. The two specifications differ in how they remove the main effect of scale: TWFE conditions on farm size linearly, whereas DML's nuisance function conditions on it nonparametrically. Because farm expenditure increases nonlinearly with scale (rising steeply among the smallest farmers before flattening as illustrated in Figure A5), and likely depends on scale jointly with other covariates, a linear control may struggle to orthogonalize scale from farm expenditures, leaving residual curvature that a nonparametric nuisance function removes.

Figures A6 and A7 illustrate the efficacy of both model specifications in orthogonalizing the outcome with respect to farm size. Each top panel plots the residuals of the two outcome models against farm size: for DML the cross-fitted nuisance residual ($\tilde{Y}_{it} = Y_{it} - \hat{n}(X_{it}, \bar{X}_i)$), and for TWFE, the residual from an analogous specification estimated without the treatment terms, $\tilde{Y}_{it} = Y_{it} - (\hat{\alpha}_i + \hat{\lambda}_t + X'_{it}\hat{\gamma})$, so that both residual series retain treatment variation and differ only in how covariates and fixed effects are partialled out. If a specification adequately captures the dependence of the outcome on farm size, then the residuals should have no curvature with respect to it. Curvature that remains admits two interpretations: either the specification has failed to fully absorb scale dependent variation in the outcome, whether directly or jointly with other covariates (functional form misspecification), or curvature is generated by variables excluded from the specification by construction (i.e. the treatment itself).

For nonfarm expenditures (Figure A6), both specifications yield distributions with best fit curves[15] that nearly perfectly align, implying both specifications effectively orthogonalize the outcome to farm scale. For farm expenditures (Figure A7), the TWFE best fit curve bends with farm size while the DML curve stays nearly flat. As a summary measure, we report the standard deviation (SD) of each fitted curve evaluated over the support, where a perfectly flat (orthogonalized) curve has an SD of zero. The SD of the DML fitted curve in Figure A7 is less than half of the TWFE specification (0.05 and 0.13, respectively). We can compare this with the near identical SD of the best fit curves in Figure A6, which are both approximately 0.02. If anything, this comparison handicaps DML since TWFE can benefit from in-sample overfitting while DML's grouped cross-fitting restricts predictions to out-of-fold observations. The flatness of the DML curve is therefore unlikely to be an artifact of an overfit learner.

The difference in residuals is not just a matter of aggregate distribution but also one of allocation. The two outcome models explain comparable aggregate variance (R^2_Y and Adj-R^2 are nearly identical in Table 3), so the divergence in residual curvature reflects how each specification explains variance across the scale dimension. The lower panels show the observation-level disagreement between model specifications ($|\tilde{Y}^{DML}_{it} - \tilde{Y}^{TWFE}_{it}| = |\hat{n}(X_{it}, \bar{X}_i) - (\hat{\alpha}_i + \hat{\lambda}_t + X'_{it}\hat{\gamma})|$). The difference in residual values illustrates the gap between the two models' fitted values: the mean value is roughly 50 percent larger, with a 36 percent larger SD, for farm expenditures than nonfarm expenditures ($\mu = 0.638$ and $\sigma = 0.535$ versus $\mu = 0.423$ and $\sigma = 0.394$, respectively). This

[15] All best fit curves in Figures A6, A7, and A8 are third-degree fitted polynomials.

further supports the intuition that the DML specification largely aligns with the TWFE specification in estimating routine nonfarm expenditures, but yields novel insight in the farm expenditure setting. Functional form therefore matters materially more on the farm margin.

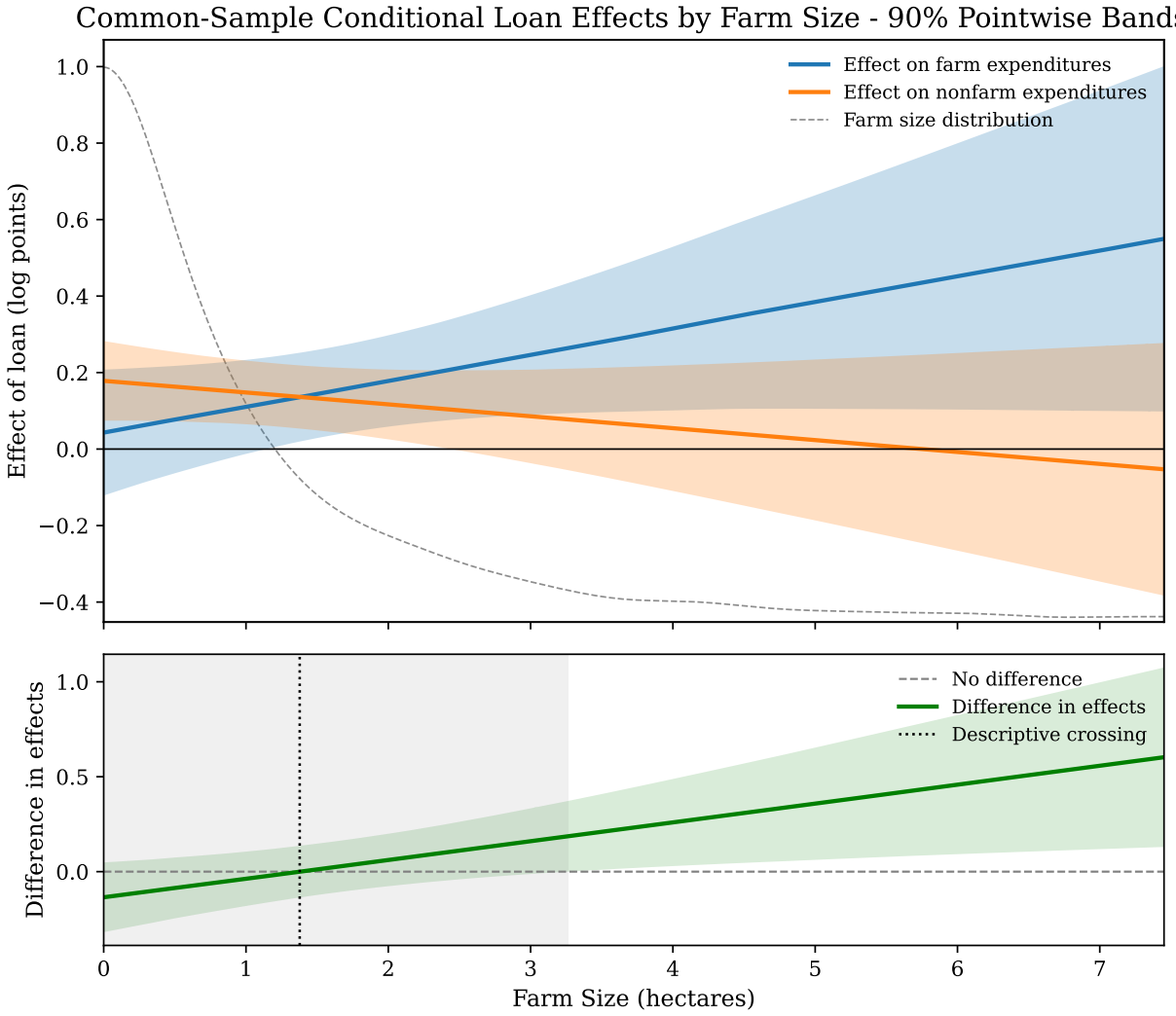
This is also why the aggregate interaction trades places between the model specifications, through two channels. First, the lower panels of Figures A6 and A7 show that the observation level disagreement between the two specifications is roughly 50 percent larger, and more dispersed on the farm margin than on the nonfarm margin. The two models are therefore genuinely fitting different expectation functions for farm expenditures, and the second stage treatment effect estimation inherits that difference. Second, the residual curvature in the upper panels identifies how the difference manifests in our point estimates: because the interaction regressor is a function of farm size, scale-correlated residual distribution that the linear control fails to absorb projects directly onto the TWFE interaction coefficient, whereas the DML nuisance function removes it before the interaction is estimated.

That said, residual curvature is not, in itself, evidence of misspecification. Both residual series deliberately retain treatment variation, so curvature generated by credit effects that vary with farm size—the heterogeneity we are investigating—is expected. Critically, however, this treatment-driven component is common to both residual series, which are constructed from the same observations, control set (X), and treatment variation. The differential in curvature (the SD of the DML curve is less than half that of TWFE, despite relying on out-of-fold predictions) therefore cannot be attributed to treatment effects, and instead indicates that the linear specification absorbs scale dependent variation less effectively. Accordingly, while we cannot determine how much of the remaining DML curvature reflects true credit effects, the comparison is consistent with a TWFE interaction coefficient that partially absorbs unmodeled scale curvature rather than treatment heterogeneity alone. We therefore treat the DML interaction estimate as imprecise rather than as a definitive null.

Descriptive crossover in expenditure responses. To explore and compare the heterogeneous treatment effects on farm and nonfarm expenditures, we replicated the analysis in Table 3 with a common estimation sample and a marginal effect plot. To do this, we restrict the nonfarm expenditure analysis to the subsample of households with positive farm expenditures ($N = 5,476$). This allows us to compare the estimated marginal effects of agricultural credit on farm and nonfarm expenditures across the same set of households. The results are reported in Table A16 and illustrated in Figure 2. The point estimates are qualitatively similar to the full sample analysis, with a positive but imprecise interaction term for farm expenditures and a negative but imprecise interaction term for nonfarm expenditures.

Figure 2 is a marginal effect plot illustrating the estimated effects of agricultural credit on farm and nonfarm expenditures over the distribution of farm sizes. The figure provides additional

descriptive support to the farm size mechanism discussed in this section: broadly the farm expenditure response rises with farm size while the nonfarm expenditure result falls. The fitted farm and nonfarm responses cross descriptively near 1.38 hectares, but their difference remains statistically indistinguishable from zero at the 10% level through approximately 3.27 hectares (the large majority of the common support).



Notes: The top panel illustrates marginal effects of agricultural credit by farm size, from the DML estimates in Table A16, with 90 percent confidence bands. The bottom papel illustrates the difference in marginal effects between farm and nonfarm expenditures, with 90 percent confidence bands.

FIGURE 2

We speculate that this descriptive pattern is consistent with credit induced routine nonfarm expenditures supporting household needs, such as food or clothing, that are more closely related to observable household characteristics, such as household size, than to farm size. By contrast, the farm expenditure response may be, in part, driven by economies of scale in farming operations, where larger farms perceive greater benefit per unit of expenditure and are better able to leverage

credit into productive farm expenditures. That said, we are unable to draw strong conclusions from this descriptive analysis, and we treat the threshold as descriptive.

Taken together, the DML and TWFE estimates suggest that receiving an agricultural loan increases both routine nonfarm expenditures and farm investment. The nonfarm expenditure response is robust across estimators and appears broadly scale invariant. The farm expenditure response is more sensitive to the estimator, but both specifications indicate that larger farms may be better positioned to leverage scale to convert credit into productive farm spending. This pattern is consistent with the interpretation that landholdings and credit access may act as complements, though the estimated effects should be interpreted cautiously given the imprecision of the DML interaction term. Consistent with this, we treat the crossing as descriptive and locate the substantive heterogeneity evidence in the input-channel analysis below.

Importantly, these heterogeneous effects are unlikely to be driven by differences in loan magnitude. As shown in Table A3, the ratio of agricultural loans to farm expenditures is higher among the smallest farms (170.6 percent in Quartile 1) than among the largest farms (57.6 percent in Quartile 4). Despite receiving loans that are large relative to their existing farm expenditures, smaller farms exhibit weaker farm investment responses. This pattern suggests that the observed heterogeneity reflects differences in expenditure allocation rather than differences in credit volume.

4.3 Sub-category analysis for farm and nonfarm expenditures

To further examine the specific mechanisms driving the effect of agricultural loans on household expenditures we perform analysis based on the components of farm expenditures (Inputs, Land, Labor, and Livestock) and nonfarm expenditures (Food, and Nonfood). Because we continue to log-transform outcomes and exclude zero expenditure observations, each subcategory is estimated on its own subsample. Results continue to reflect *quasi-intensive-margin* responses among households with positive expenditures. However, because zero-valued expenditures are substantially more prevalent among disaggregated farm expenditure outcomes discussed in Section 3.3, point estimates may be more affected by treatment induced extensive margin entries and may therefore diverge more from the pure intensive margin estimand. TWFE results are reported in Appendix A, Tables A6 and A7.

Table 4 presents the disaggregated farm expenditure treatment effects. For input, land, and labor expenditures there are no statistically significant results. This is likely a result of heterogeneity in investment strategies across households (i.e. some households intensify through input expenditures such as fertilizer utilization, while others hire labor or acquire land) and reduced power due to smaller sample sizes. However, coefficient signs are universally positive, and receiving agricultural loans increases livestock expenditures by 43.5 percent (0.361 log points), significant at the 10 percent level. Given that this category includes recurring costs such as feed and healthcare,

this pattern is consistent with credit supporting ongoing capital maintenance rather than extensive margin herd expansion[16]. These results are consistent in sign with the TWFE specification in Table A6, although the DML specification notably differs in magnitudes. However, the null result on disaggregated input expenditures (Column 1) may obscure specific treatment effects, so we turn from input spending to a physical measure of input use, fertilizer application intensity, observed on the positive use subsample (N=1,676, against 3,670 for input expenditures).

TABLE 4.
Heterogeneous Effects of Agricultural Loans on Disaggregated Farm Expenditures

	Input Exp. (log) (1)	Land Exp. (log) (2)	Labor Exp. (log) (3)	Livestock Exp. (log) (4)	Fertilizer (log kg/ha) (5)
Received Agricultural Loan (Loan)	0.044	0.094	0.055	0.361*	0.009
	(0.098)	(0.220)	(0.082)	(0.204)	(0.115)
Loan × Farming land area (HA)	0.067	0.035	0.037	0.032	0.189***
	(0.062)	(0.184)	(0.049)	(0.087)	(0.070)
Specification					
Controls	Yes	Yes	Yes	Yes	Yes
Federal State FEs	Yes	Yes	Yes	Yes	Yes
Survey Wave FEs	Yes	Yes	Yes	Yes	Yes
Household CREs	Yes	Yes	Yes	Yes	Yes
Learners					
Loan $\sim X$	PLH-L	PLH-L	PLH-L	PLH-L	PLH-L
Expenditures $\sim X$	ERH-R	ERH-R	ERH-R	ERH-R	ERH-R
K-folds	5	5	5	5	5
Diagnostics					
R^2_Y (Outcome: $Y \sim X$)	0.362	0.234	0.238	0.226	0.387
AUC (Loan $\sim X$)	0.599	0.656	0.616	0.626	0.596
AP (Loan $\sim X$)	0.076	0.121	0.087	0.103	0.082
Observations	3,670	352	4,010	1,176	1,676

Standard errors in parentheses. Double Machine Learning (PLR), $n_{\text{rep}} = 30$.

R^2_Y reports nuisance learner performance for the outcome model.

AUC (Area Under Curve) and AP (Average Precision) report nuisance learner performance for the treatment model.

ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.

PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked with a Logistic meta-learner.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

To test whether the size-dependent investment effects observed in our heterogeneous effects model (Table 3) are driven by agricultural intensification, we specifically examine fertilizer appli-

[16]The one component where the linear benchmark is the more precise of the two is land (Table A6: 0.422, P-value=0.101). We do not interpret this divergence in either direction, as this specification is by a large margin the least powered analysis with $N = 352$.

cation intensity (kg/ha) in Column 5. The results suggest that fertilizer intensification is important for larger smallholders who receive credit. The coefficient (0.189 log points) on the interaction term is positive and statistically significant at the 1 percent level, indicating that each additional hectare increases the credit effect on fertilizer intensity by 20.8 percentage points. Of particular note, this positive estimate runs opposite to the descriptive gradient we observe in Table A3, where the smallest farms utilize fertilizer more intensively (Quartile 1: 858 kg/ha) than the largest farms (Quartile 4: 392 kg/ha), easing concern that the gradient simply reflects the association between farm size and fertilizer use. Similarly, Table A6 differs in its point estimates, indicating a large insignificant ATE and a small insignificant HTE. This pattern is consistent with a nonlinear fertilizer investment decision that the TWFE specification does not recover.

We again interpret the two specifications against one another through their residual distributions (Figure A8), applying the diagnostics developed in Section 4.2. The same pattern observed for total farm expenditures is even more pronounced for fertilizer utilization. The fertilizer fitted curve is close to flat under DML but visibly curved under TWFE, and the gap between model specifications is even larger: the SD of the fitted curve across the support is 0.04 for DML versus 0.18 for TWFE, against 0.05 and 0.13 for total farm expenditures. This places the DML residual near the benchmark observed where both specifications agree (0.02 on the nonfarm margin), while the TWFE value is the largest in our analysis. As in Section 4.2, the loan's contribution to residual curvature is identical across the two series, so the widened differential cannot reflect credit effects. Instead, the gap indicates the linear specification absorbs scale dependent variation in fertilizer use even less completely than in aggregate farm spending. The lower panel corroborates the magnitude of the model disagreement: the observation-level gap between model residuals (equivalent to the gap in predictions) is larger for fertilizer than for aggregate farm expenditures ($\mu = 0.762$ and $\sigma = 0.595$ versus $\mu = 0.638$ and $\sigma = 0.535$, respectively).

Taken together, two features make this our central evidence of scale dependent investment. First, the average effect is essentially zero (0.009, P-value=0.937) while the interaction is strong, so the result is not that credit raises fertilizer use on average but that it does so only as scale rises. Second, the gradient is recovered by DML but absent under TWFE (see Table A6: -0.005, P-value=0.946). This is consistent with our residual analysis above (Figure A8), where the DML nonparametric estimation more thoroughly removes farm size driven variation in fertilizer utilization, leaving the interaction to be estimated against a residual purged of the scale curvature that contaminates its linear counterpart. Moreover, it is unlikely that an overfit learner would reproduce the theoretically expected sign, at high significance, precisely at the relevant margin. Indeed, the coherence of the fertilizer utilization result with the preceding mechanism is what leads us to interpret the total farm expenditure divergence as a signal rather than instability induced by rare treatment.

We also replicated the analysis using two additional physical-input measures: total hired

person-days and herbicide application. Neither analysis produced statistically significant results; coefficients were positive for the former and mixed in sign for the latter. For brevity, these results are not reported but are available for reproduction.

TABLE 5.
Heterogeneous Effects of Agricultural Loans on Nonfarm Expenditures

	Food Expenditures (log)	Non-Food Expenditures (log)
Received Agricultural Loan (Loan)	0.155***	0.065*
	(0.060)	(0.035)
Loan × Farming land area (HA)	−0.055	0.029
	(0.038)	(0.026)
Specification		
Controls	Yes	Yes
Federal State FEs	Yes	Yes
Survey Wave FEs	Yes	Yes
Household CREs	Yes	Yes
Learners		
Loan $\sim X$	PLH-L	PLH-L
Expenditures $\sim X$	ERH-R	ERH-R
K-folds	5	5
Diagnostics		
R^2_Y (Outcome: $Y \sim X$)	0.335	0.604
AUC (Loan $\sim X$)	0.625	0.625
AP (Loan $\sim X$)	0.079	0.079
Observations	5,898	5,900

Standard errors in parentheses. Double Machine Learning (PLR), $n_{\text{rep}} = 30$.
R^2_Y reports nuisance learner performance for the outcome model.
AUC (Area Under Curve) and AP (Average Precision) report nuisance learner performance for the treatment model.
ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.
PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked with a Logistic meta-learner.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5 presents treatment effects for the components of nonfarm expenditures. Receiving an agricultural loan increases food expenditures by 16.7 percent (0.154 log points) and non-food expenditures by 6.7 percent (0.065 log points), statistically significant at the 1 and 10 percent levels respectively. The interaction terms with farm size are statistically insignificant, consistent with the earlier finding that nonfarm expenditure responses do not vary systematically with landholdings. These results closely align with our TWFE specification in Table A7, consistent with our earlier assertions that agricultural credit broadly relaxes constraints on routine expenditures in a relatively mechanical fashion, with limited benefit from nonparametric estimation.

Although the estimated effects on food expenditures are sizable, they should not be inter-

preted as a one-to-one mapping from loan value to consumption. As shown in Table A3, the ratio of mean agricultural loan value to total nonfarm expenditures is relatively small (4.4–6.2 percent across quartiles). The estimated percentage increases therefore likely reflect broader household reallocation following liquidity relief, rather than direct spending of loan proceeds. In this sense, credit access appears to relax short-run budget constraints, allowing households to adjust consumption patterns alongside farm investment decisions.

4.4 Robustness Checks

4.4.1 Sensitivity to cross-fitting folds

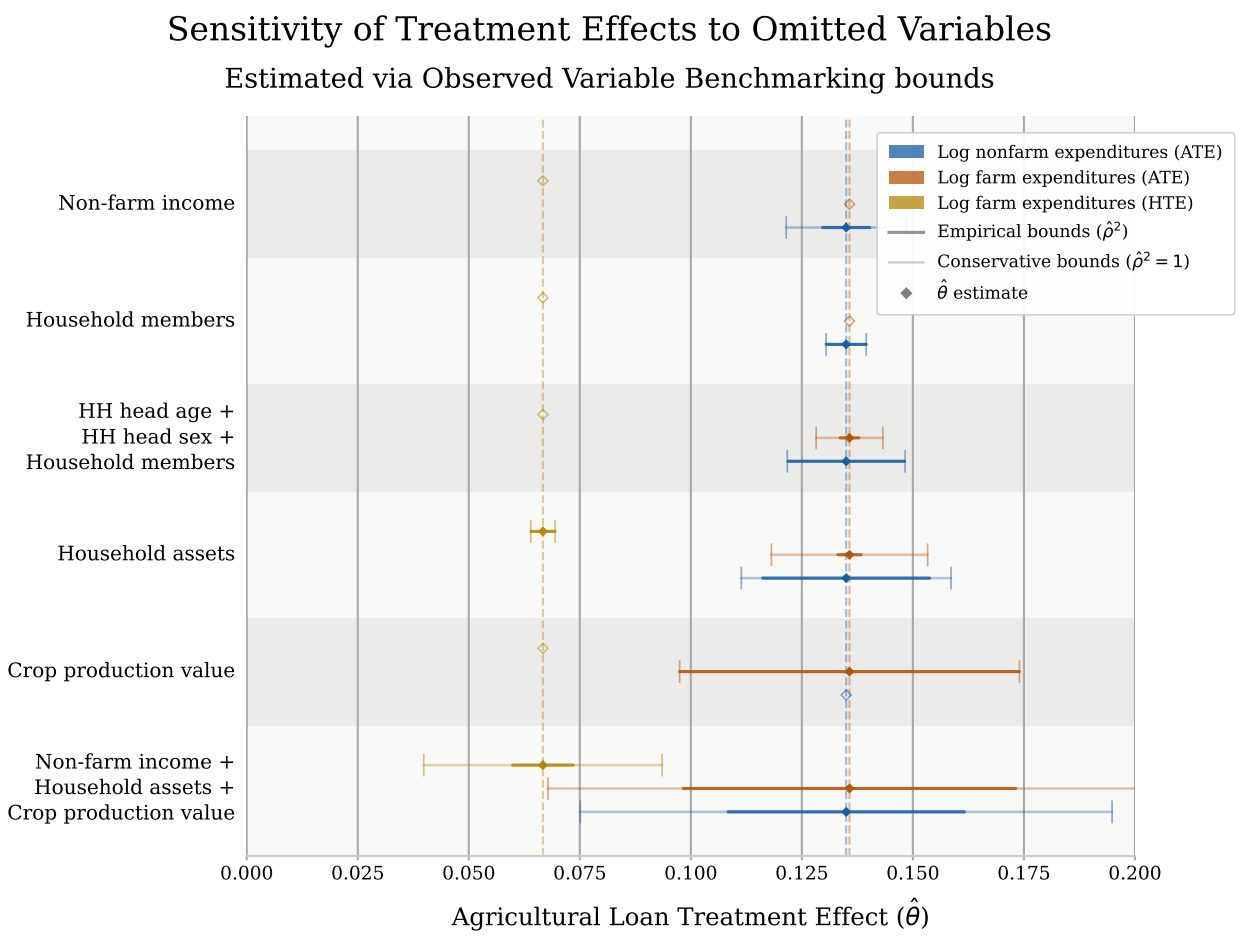
We assess the stability of our core estimates to model specification. Specifically, DML results can be sensitive to the random partitioning of data during the cross-fitting process. Tables A8 and A9 demonstrate the sensitivity of our findings to model design by varying the number of cross-fitting folds ($K \in \{3, 5, 8\}$). The results indicate that our estimates are highly stable across fold specification. The greatest sensitivity is observed in Table A9, with a difference of approximately 0.005 between the 3-fold and 8-fold specifications. The majority of point estimates differ by less than 0.003 across fold specifications.

4.4.2 Robustness to Omitted Variable Bias

Approximately 76 percent of agricultural loans in our sample originated from informal or semi-formal sources (e.g. friends, money-lenders, savings associations). As a result, selection into credit is tied to opaque social factors. Consequently, unobservable confounders such as reputation, social networks, and entrepreneurial spirit represent a challenge to identification. This motivated our use of DML augmented for panel data, controlling for time-invariant unobservables with Mundlak means, and complex nonparametric relationships with our learners. The former design choice has been observed to significantly enhance performance over traditional linear TWFE specifications (Fuhr and Papies, 2024), while the latter has proven specifically useful in the context of noisy LMIC credit markets (Delavallade and Godlonton, 2023).

Nevertheless, to address the threat of time-varying omitted variable bias (OVB) we perform a sensitivity analysis of our results following Chernozhukov et al. (2026). Specifically, we benchmark our treatment effect estimates against pseudo-OVB estimated on observables. We do this by sequentially removing observables from our control vector X, measuring the impact on the predictive power of our nuisance functions, and estimating a maximally adversarial (i.e. perfectly correlated confounding between treatment and outcome) confidence bound.

Figure 3 illustrates the confidence bounds of our point estimates over the six most impactful omitted variable combinations. We focus our discussion around the benchmarked impact of removing our core endowment variables: nonfarm income, household assets, and crop production



Notes: Treatment effect bounds after omitting the indicated controls, following the observed-variable benchmarking approach of Chernozhukov et al. (2026). Empirical bounds use estimated confounding strength ($\hat{\rho}^2$); conservative bounds assume maximal confounding ($\hat{\rho}^2 = 1$).

FIGURE 3

value. Removing these principal endowment variables shifts our treatment effect estimates toward zero by up to 44.4 percent for nonfarm expenditures, 50.0 percent for farm expenditures, and 40.2 percent for the farm expenditures interaction term.

This bounding exercise indicates that an unobserved confounder would need to be approximately 2 to 2.5 times as impactful on both credit access and expenditures as the combined effect of our core endowment variables to drive our point estimates to zero. More conservatively, an unobserved confounder would need to exert roughly the same magnitude of influence as our core endowment variables to render our nonfarm expenditure ATE point estimate statistically insignificant at the 10 percent level, and about one-quarter the magnitude to do the same for the farm expenditure ATE point estimate.

While soft variables such as reputation play an important role in accessing credit in informal settings, the magnitude of time-varying confounding necessary to nullify our results appears im-

plausibly large. Regarding entrepreneurial drive, household expenditures remain fundamentally bounded by available assets and liquidity. Because soft variables are unlikely to dictate aggregate spending to the degree of fundamental economic indicators, we conclude that our findings are robust to unobserved confounding.

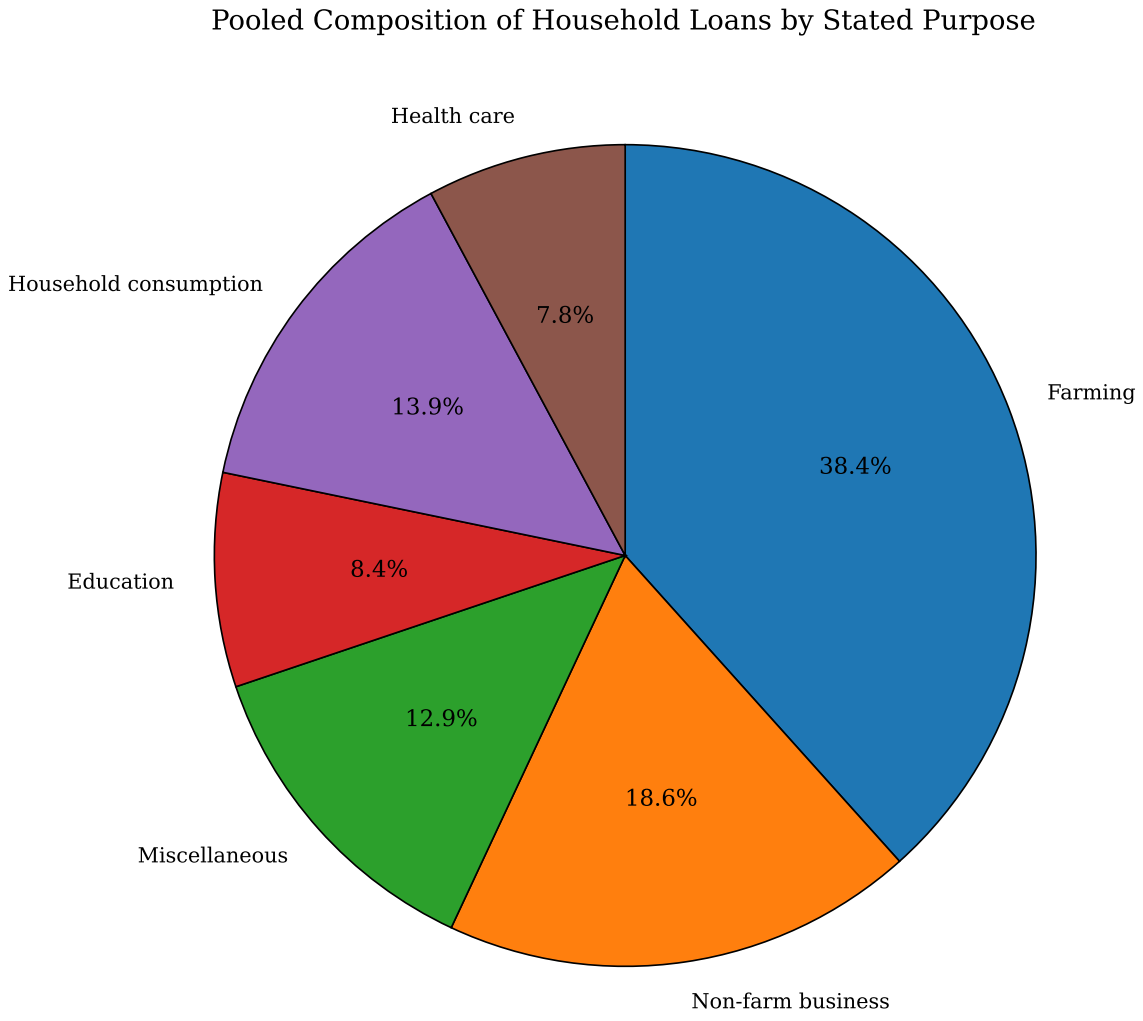
4.4.3 Sensitivity to Loan Specification

To identify agricultural loans, we use recipient self reported loan purpose for formal and informal sources to maximize the number of treated observations (see Section 2.3). Even still, treatment remains very rare in our sample, with approximately 4.9% of households receiving agricultural credit across both survey waves. In this section, we discuss the effect of receiving any loan on household expenditures. Figure 4 illustrates that agricultural loans constitute the plurality of credit received by smallholders in our sample. Expanding our criteria to include any loan received by a household increases the treated share from approximately 4.9% to 11.8%.

Tables A10 and A11 report the expanded analysis. Under the broadened treatment, the DML ATE on farm expenditures is positive but statistically insignificant (0.057, P-value=0.228), as is the effect on nonfarm expenditures (0.050, P-value=0.112). Both point estimates retain the sign and approximate structure of our core specification (Table 3) but are smaller and no longer distinguishable from zero. The TWFE specification largely mirrors the DML point estimates and are similarly insignificant in both cases (0.052, P-value=0.435 for farm and 0.073, P-value=0.103 for nonfarm expenditures).

The core estimates use agricultural loans alone, while the broadened treatment adds credit earmarked for other purposes (education, health care, or nonfarm businesses), roughly doubling the treated share from 4.9 to 11.8 percent. The shrinkage and loss of significance under the broader definition is consistent with our core results being specific to agricultural credit rather than a general feature of household borrowing. However, despite the notable change in coefficient magnitudes, the point estimates are not statistically distinguishable at traditional levels. For example, take the largest change, the nonfarm ATE: the difference in coefficients is ~ 0.084 ($= 0.134 - 0.050$), with a pooled standard error of ~ 0.059 ($= \sqrt{0.049^2 + 0.032^2}$), implying $t \approx 1.44$ (P-value=0.151). This, if anything, is a conservative estimate given that we would expect the two coefficients to be correlated, so we do not read it as evidence of a behaviorally distinct response to non-agricultural credit.

Our heterogeneous results follow the same pattern. The DML interaction coefficient for farm expenditures is positive but insignificant (0.031, P-value=0.342), and the nonfarm interaction is small and insignificant (-0.001, P-value=0.975). The TWFE specification again diverges from the DML on the farm margin, with a positive and significant farm interaction (0.095, P-value=0.044) alongside an insignificant nonfarm interaction. That the divergence observed in the core analysis



Notes: Loans pooled across waves by recipient stated purpose, and the sample remains limited to paired agricultural households.

FIGURE 4

(Table 3) survives a different treatment definition supports our earlier analysis that farm margin disagreement reflects functional form rather than the rare agricultural loan treatment. Broadening the treatment therefore preserves the sign and qualitative structure of our core estimates while reducing their magnitude and precision—supporting our headline findings being specific to agricultural credit rather than broader relaxation of liquidity constraints.

4.4.4 Effect of Extension Services on Expenditures

A key challenge in our analysis is the sparsity of agricultural credit, with approximately 4.9% of households receiving agricultural loans when pooled across both survey waves. This raises the concern that our results may be driven by a small number of treated households, and that

the observed treatment response may reflect unobserved characteristics of these households. As a complementary comparison using a more prevalent treatment, in this section we examine the effect of receiving agricultural extension services on household expenditures. For this analysis we utilize responses to the Agricultural Questionnaire, specifically modules 11L and A5B, which ask households whether they received agricultural extension services in the past 12 months or the preceding agricultural year in the post-planting (PP) and post-harvest (PH) surveys respectively.

In each survey wave and survey module, respondents answer questions regarding whether extension services were received, topics of discussion with extension agents (e.g. new seed varieties or weather event risks), and the source of the extension services (e.g. government, NGO, peer farmer). Similarly to our agricultural credit analysis, we define treatment as a binary indicator for whether a household received any extension services. Additionally, we focus our analysis on the PP survey module (11L) as it is more closely aligned with the timing of our agricultural credit data (both collected in the PP survey and indicate treatment in the preceding 12 months). For completeness we also report results from the pooled PP and PH survey modules. Our treatment variables are significantly more common than agricultural credit, with approximately 16.4 and 31.4 percent of households in our sample receiving extension services under the PP only and combined PP/PH survey modules respectively.[17] Besides the treatment definition, the analysis follows the same DML and TWFE specifications as our core agricultural credit analysis.[18]

The results of our extension service analysis (reported in Appendix A, Tables A17 and A18) align with our core agricultural credit analysis. For farm expenditures, our baseline results are positive and statistically significant under both DML (0.121, P-value=0.002) and TWFE (0.146, P-value=0.011), indicating that extension services have a similar effect on baseline farm expenditures as agricultural credit (Table 2). A similar pattern of estimator disagreement appears in the heterogeneous response where the TWFE result is large and statistically significant (-0.105, P-value=0.001), with the key difference of a negative sign. Conversely, the DML result is small in magnitude, positive, and imprecise (0.009, P-value=0.70). Notably, although the DML heterogeneous result remains positive, the imprecise estimate is significantly smaller in magnitude than the core analysis (0.066, Table 3). Together, our results continue to illustrate the DML specification producing novel estimates relative to its TWFE benchmark, with both signs and magnitudes of coefficients changing between the two specifications.

The nonfarm expenditure responses to post-planting extension services are similar across both

[17] Specifically, in the PP period 17.7% and 15.1% of households received extension services in Wave 4 and 5 respectively. Pooling across the PP and PH surveys, 30.9% and 31.9% of households received extension services in Wave 4 and 5 respectively

[18] During the course of our analysis, we observed that one hyperparameter utilized in the logistic meta-learner significantly degraded the performance of our treatment propensity nuisance function modeling. Specifically, the internal cross-validation hyperparameter was removed for this analysis. The change has no impact on the DML or cross-fitting methodologies outlined in 3.2, but should be noted as the sole difference in model specification between the core and extension analyses.

DML and TWFE specifications, with no baseline or heterogeneous statistically significant coefficients. These results contrast our findings in the core agricultural credit analysis, where nonfarm expenditure responses were positive and statistically significant. The finding that extension services increase farm expenditures while producing no statistically detectable effect on nonfarm expenditures may illustrate that extension services and credit influence behavior via different channels.

Taken together, we can draw three useful conclusions from this robustness check. First, the positive baseline farm expenditure finding persists under a distinct agricultural support treatment with a substantially larger share of observations receiving treatment than agricultural credit. Second, the information provided by extension services may encourage agricultural intensification through a different channel from the direct liquidity relief provided by credit. The weaker estimated nonfarm response is consistent with this interpretation, although indirect consumption effects through agricultural intensification remain possible. Finally, the DML and TWFE specifications continue to produce similar nonfarm expenditure estimates, while differing more noticeably in the estimated farm size interaction. The reoccurrence of this pattern across two distinct treatments with different treatment prevalence suggests that the disagreement is not confined to the sparse agricultural credit setting.

5 Discussion and Limitations

This paper examines how agricultural credit shapes household expenditure allocation in Nigeria and how these responses vary with farm scale. The results indicate that credit raises both farm and nonfarm expenditures, and that the two responses differ in how they relate to farm scale. The farm investment response is concentrated among larger landholders and operates through fertilizer utilization, whereas the nonfarm expenditure response is broadly invariant to scale. These patterns align with models of production under incomplete markets, where the returns to purchased inputs depend on scale and complementary constraints (Carter and Olinto, 2003; Feder, 1985).

For relatively larger smallholders, agricultural credit is associated with stronger farm investment responses, identified through the fertilizer intensity channel. This is consistent with credit and landholdings acting as complements: where scale permits profitable intensification, credit can facilitate productivity-enhancing input use.

By comparison, the scale insensitive nonfarm routine expenditure response is itself informative. The effect on nonfarm expenditures is positive but shows no evidence of varying with farm size (the interaction coefficient is small, negative, and statistically indistinguishable from zero under both DML and TWFE specifications across the land distribution). Whereas the farm investment response is conditional on production capacity (proxied by farm size), the nonfarm con-

sumption response is not, which points to credit relaxing a general household liquidity constraint rather than a farm specific one. The two channels are therefore distinguished by their relationship to scale: flat for consumption, increasing for farm investment. It is this divergence, more than the average treatment effect, that the expenditure allocation approach recovers.

The pattern also sharpens the interpretation of smaller farms. Smaller farms realize a similar routine consumption response to that of larger households but either face lower perceived returns to credit-induced intensification or more tightly binding liquidity constraints for consumption. As a result, smaller landholders proportionally invest less when receiving credit. This is consistent with the larger agricultural finance and microcredit literature showing that credit impacts are highly heterogeneous and shaped by underlying production opportunities and risk environments (Karlan et al., 2014; Hussain and Thapa, 2016; Meager, 2019).

From a food systems perspective, the results highlight the dual role of agricultural credit. For some households, credit supports productive investment and potential yield gains; for others, it primarily eases short-run liquidity constraints. Evaluating agricultural finance policies solely on average production responses may therefore obscure important welfare and resilience effects. Designing credit instruments that account for farm heterogeneity may improve alignment between policy objectives and household behavior.

Several limitations warrant consideration. First, although the DML framework controls for high-dimensional observed confounding and incorporates correlated random effects to control for time-invariant unobserved confounding, identification relies on the assumption of conditional unconfoundedness. This is particularly relevant in settings where informal lending dominates and selection may depend on unobserved factors such as reputation or social networks. While the sensitivity analysis in Section 4.4.2 supports the robustness of our results to unobserved confounding, our identification strategy ultimately relies upon observables and cannot provide the same degree of causal certainty as a randomized design. Second, loan magnitudes are weakly correlated with land endowments in our sample, suggesting that credit constraints remain binding across the farm size distribution. Consequently, the results reflect a context of pervasive credit rationing and may not generalize to environments with deeper formal financial penetration.

An additional limitation concerns measurement of food expenditures. Our food expenditure data are constructed from annualized 7-day recall data, which may introduce aggregation error and inflate expenditure levels relative to observed income streams. While our analysis focuses on percentage changes rather than levels, any systematic measurement error in food reporting could effect the magnitude of estimated effects. We therefore interpret the food expenditure results as indicative of directional responses rather than precise level shifts.

Finally, the estimated percentage effects on food expenditures are large relative to mean loan size. This pattern suggests that credit access may induce broader expenditure reallocation rather than direct mechanical spending of loan proceeds. However, the data do not permit direct ob-

servation of intra-household allocation of loan funds, and we therefore interpret these responses as equilibrium adjustments following liquidity relaxation rather than as precise mappings from credit to specific expenditures.

6 Conclusions

This paper investigates how agricultural credit affects expenditure allocation among smallholder farmers in Nigeria and how these effects vary with farm scale. Using panel-adjusted Double Machine Learning, we estimate both average and heterogeneous treatment effects while accounting for high-dimensional confounding and time-invariant heterogeneity (Chernozhukov et al., 2018, 2025; Semenova and Chernozhukov, 2021).

Our findings indicate that agricultural credit raises both farm and nonfarm expenditures, and that the two responses differ in how they relate to landholdings. For relatively larger smallholders, credit is associated with stronger farm investment responses, identified through fertilizer intensification. The nonfarm routine expenditure response, by contrast, is positive but broadly invariant to farm size, pointing to relief of a general household liquidity constraint rather than a farm specific one. For smaller farms, expenditure adjustments are more concentrated in nonfarm categories. Specifically, smaller farms realize the common nonfarm routine consumption response, but, whether because of lower perceived returns to scale or more tightly binding liquidity constraints, do not exhibit the scale conditional investment response. These results are consistent with models of agricultural production under incomplete markets, where the profitability of purchased inputs depends on scale, supervision, and complementary constraints (Carter and Olinto, 2003; Feder, 1985). They also align with evidence from the broader microcredit and agricultural finance literature showing that credit functions as part of an integrated household portfolio rather than as project-specific capital (Angelucci et al., 2015; Banerjee et al., 2015; Meager, 2019).

By documenting continuous heterogeneity across the farm size distribution, this study helps reconcile mixed findings in prior work. Experimental and quasi-experimental studies have found positive investment responses when complementary conditions—such as insurance or favorable timing—are present (Karlan et al., 2014; Burke et al., 2019; Fink et al., 2020). In settings characterized by pervasive credit rationing and exposure to shocks, however, liquidity relief may primarily facilitate routine consumption or risk management. Our Nigerian case reflects this duality by showing that credit also acts as a complementary condition for agricultural investment but most intensely for larger smallholders.

Several caveats remain. Identification relies on the assumption of conditional unconfoundedness given a rich control vector and correlated random effects structure. Although the DML framework mitigates overfitting and high-dimensional bias (Belloni et al., 2014; Chernozhukov et al., 2018), informal lending environments may involve unobserved selection factors such as

reputation or social ties. In addition, loan magnitudes are only weakly correlated with land endowments in our data, suggesting that credit constraints bind across the farm size distribution. The results therefore characterize behavior under severe credit rationing and may not generalize to contexts with deeper formal financial markets.

More broadly, the picture that emerges is of a single, fungible loan that households deploy according to both their consumption needs and agricultural investment capacity. It may support productive intensification where scale permits, while simultaneously enhancing resilience and consumption stability among more constrained farms. Designing financial instruments that recognize this heterogeneity, rather than targeting uniform production responses, may be central to advancing effective and equitable food system policy in low-income settings.

6.1 Policy and Methodological Implications

The findings have several implications for the design and evaluation of agricultural credit programs. First, the heterogeneous expenditure responses suggest that uniform loan products are unlikely to generate homogeneous outcomes. Credit appears to support productive intensification primarily among relatively larger smallholders, while functioning as short-run liquidity support among smaller farms. Credit design that accounts for farm scale (through differentiated loan sizes, repayment structures, or complementary input-linked arrangements) may better align financial instruments with production capacity and household constraints.

Second, the results are directly relevant to large-scale public lending initiatives in Nigeria, including NIRSAL and the Anchor Borrowers' Programme (ABP). These programs operate in environments characterized by severe credit rationing and heterogeneous farm structures. Our evidence suggests that scale-sensitive guarantees, repayment structures, and targeting mechanisms may improve program effectiveness. For smaller farms, credit may primarily enhance resilience and consumption stability; for relatively larger smallholders, it may facilitate input intensification and yield growth. Recognizing these differentiated channels may help refine eligibility criteria, loan terms, and complementary services within such programs.

Third, the findings highlight the importance of integrating credit with complementary institutions. In settings where risk exposure and input access constraints remain binding, liquidity alone may not translate into sustained productivity gains among smaller farms. Prior experimental evidence emphasizes the role of insurance, savings mechanisms, and seasonal timing in shaping investment responses (Karlan et al., 2014; Burke et al., 2019; Fink et al., 2020). Aligning agricultural credit programs with risk-sharing mechanisms and input markets may therefore enhance their impact.

Methodologically, this study illustrates both the value and the limits of flexible estimation in our setting. Benchmarked against analogous TWFE specifications, DML converges with the linear

model on routine nonfarm expenditures and diverges on the farm investment margin. On the farm investment margin, expenditure is nonlinear in scale, and the agricultural intensification signal is only recovered through nonparametric nuisance estimation. The value of flexibility is therefore localized to contexts where the response function exhibits a high degree of complexity. This is informative to researchers: DML presents an advantageous estimation strategy for recovering treatment effects from observational datasets, but its benefit beyond traditional econometric tools is context dependent and can be diagnosed with relatively simple residual diagnostics.

More broadly, the evidence underscores that agricultural credit serves multiple functions within rural economies. Evaluations focused solely on average production effects risk overlooking welfare and resilience gains among households for whom credit primarily relaxes liquidity constraints. Designing financial architectures that acknowledge this heterogeneity is central to advancing effective and sustainable food system policy.

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Appendix A

TABLE A1.
Balance Table

Variable	Did Not Receive Agricultural Loan (N ≈ 5,604) Mean (SE)	Received Agricultural Loan (N ≈ 296) Mean (SE)	Difference (1) - (2) Diff.
Farming Land Area (Hectares)	1.184 (0.019)	1.345 (0.085)	-0.161*
Value of Farm Production (1,000s ₦)	354.15 (7.61)	445.92 (36.78)	-91.768**
Assets (1,000s ₦)	270.21 (7.19)	228.16 (26.14)	42.048
Nonfarm Income (1,000s ₦)	330.09 (7.67)	377.97 (39.69)	-47.879
Livestock Holdings (#)	0.772 (0.024)	0.672 (0.084)	0.100
Has Formal Land Rights (binary 0/1)	0.137 (0.005)	0.193 (0.024)	-0.056**
Income Shock (#)	0.200 (0.007)	0.176 (0.026)	0.025
Production Shock (#)	0.962 (0.016)	1.078 (0.071)	-0.116
Price Shock (#)	1.069 (0.018)	1.108 (0.083)	-0.039
Head Marital Status (binary 0/1)	0.776 (0.006)	0.817 (0.023)	-0.041*
Head Age	51.745 (0.199)	48.990 (0.813)	2.755***
Head Sex (binary 1 ≡ Male)	0.815 (0.005)	0.845 (0.021)	-0.029
Number of Members in the Household	5.940 (0.046)	6.409 (0.184)	-0.469**
Number of Adult Members in the Household	2.932 (0.020)	2.959 (0.083)	-0.027
Household Has Phone Access (binary 0/1)	0.931 (0.003)	0.943 (0.014)	-0.011
Household Has Internet Access (binary 0/1)	0.349 (0.006)	0.365 (0.028)	-0.016
Probability of Moderate Food Insecurity	0.456 (0.006)	0.442 (0.025)	0.013
Food Consumption Score	48.618 (0.234)	50.243 (1.087)	-1.626
Received Non-Agricultural Loan (binary 0/1)	0.083 (0.004)	0.078 (0.016)	0.006

Notes: Standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Includes all observations in baseline and heterogeneous Nonfarm Expenditure models (i.e. observations with non-missing farm size values).

Observation counts vary slightly due to missing data.

TABLE A2.
Description of Key Variables

Treatment Variable	Description
Received Agricultural Loan (Binary 0/1)	Indicator variable, true when the household received a loan with a stated purpose for farm expenditures.
Outcome Variables	
Input Expenditures (1,000s ₦)[a]	Seeds, fertilizer, pesticides, machinery/tools.
Land Expenditures (1,000s ₦)[a]	Land purchases and rent.
Labor Expenditures (1,000s ₦)[a]	Hired labor, rented livestock labor.
Livestock Expenditures (1,000s ₦)[a]	Livestock purchases, livestock upkeep costs.
Farm Expenditures (1,000s ₦)	Sum of: Input, Land, Labor, and Livestock Expenditures.
Food Expenditures (1,000s ₦)[a]	Imputed value of food consumption, the product of purchased food consumed and respondent recalled market price. Averaged between post-planting and post-harvest surveys.
Non-Food Expenditures (1,000s ₦)[a]	Household expenditures on goods (durable and non-durable, e.g. cigarettes, appliances), and services (e.g. vehicle repairs, insurance). **Excludes:** nonfarm business, education, health, debt, rent, vehicles, ceremonies, and holidays.
Nonfarm Expenditures (1,000s ₦)	Sum of Food and Non-Food Expenditures: intended to represent average daily consumption while excluding lumpy irregular consumption events.
Select Control Variables	
Farming Land Area (Hectares)[a,b]	Size in hectares of land owned by household, includes actively harvested, fallow fields, and pasture land.
Value of Farm Production (1,000s ₦)[a,b]	Value of all crop production produced by a household, constructed from production statistics multiplied by applicable crop prices by region.
Assets (1,000s ₦)[a,b]	Sum of the estimated value of durable goods owned by a household (excluding dwelling and agricultural plots).
Nonfarm Income (1,000s ₦)[a,b]	Income from nonfarm businesses, remittances, and all other sources.
Livestock Holdings (#)[b]	The Tropical Livestock Unit value for a household in the post-planting survey. Provides a standardized measure of livestock density including all recorded livestock such as cows, sheep, and chickens.
Has Formal Land Rights (Binary 0/1)[b]	Related to agricultural plots, the household has documentation of land rights (at least one plot).
Head Marital Status (Binary 0/1)	Indicates if the head of the household is currently married, all other classifications such as divorced, separated, or widowed are classed as zero.
Food Consumption Score (#)	The Food Consumption Score (FCS) is an index that measures the diversity and frequency of food groups consumed, which is then weighted according to the relative nutritional value of the consumed food groups (ranges 0 to 100).
Received Non-Agricultural Loan (Binary 0/1)	Indicator variable, true when the household received a loan with a stated purpose besides farm expenditures.

[a] Winsorized to top 1%
[b] Imputation performed by Evans School Policy Analysis and Research Group (UW EPAR, 2025).

TABLE A3.
Summary Statistics by Farm Size Quartile

	Mean	Mean by Farm Size Quartiles			
	Full Sample	Q1	Q2	Q3	Q4
Agricultural Loan Variables					
Received Agricultural Loan (Binary 0/1)	0.05	0.04	0.04	0.05	0.06
Size of Agricultural Loan (1,000s ₦) (Recipients Only)	102.85	96.75	110.20	82.34	108.77
Outcome Variables					
Input Expenditures (1,000s ₦)[a]	33.20	14.55	30.58	39.37	55.25
Land Expenditures (1,000s ₦)[a]	4.29	2.63	4.32	4.38	6.32
Labor Expenditures (1,000s ₦)[a]	64.56	31.55	58.17	66.04	102.44
Livestock Expenditures (1,000s ₦)[a]	32.31	19.07	36.16	35.20	34.80
Farm Expenditures (1,000s ₦)	117.98	56.71	112.89	131.52	188.76
Food Expenditures (1,000s ₦)[a]	1,538.36	1,660.10	1,569.41	1,484.12	1,351.75
Non-Food Expenditures (1,000s ₦)[a]	427.67	445.44	424.95	402.94	414.76
Nonfarm Expenditures (1,000s ₦)	1,966.03	2,105.53	1,994.36	1,887.06	1,766.51
All Fertilizer Use (kg/ha)	753.05	857.96	1,174.69	745.21	391.82
Ratios					
Agricultural Loan to Total Farm Expenditures	87.2%	170.6%	97.6%	62.6%	57.6%
Agricultural Loan to Nonfarm Expenditures	5.2%	4.6%	5.5%	4.4%	6.2%
Control Variables					
Farming Land Area (Hectares)[a,b]	1.18	0.11	0.43	1.03	3.16
Value of Farm Production (1,000s ₦)[a,b]	355.75	88.01	250.43	405.45	679.10
Assets (1,000s ₦)[a,b]	276.68	316.97	259.25	236.84	280.37
Nonfarm Income (1,000s ₦)[a,b]	349.87	367.59	361.43	322.12	299.64
Livestock Holdings (#)[a,b]	0.79	0.21	0.45	0.82	1.56
Has Formal Land Rights (binary 0/1)[a,b]	0.14	0.11	0.12	0.14	0.19
Income Shock (#)	0.20	0.20	0.18	0.21	0.21
Production Shock (#)	0.95	1.00	0.99	0.97	0.89
Price Shock (#)	1.07	1.33	1.14	1.04	0.78
Head Marital Status (binary 0/1)[c]	0.77	0.60	0.78	0.83	0.89
Head Age[c]	51.74	55.56	51.17	49.86	49.96
Head Sex (binary 1 ≡ Male)[c]	0.81	0.65	0.83	0.86	0.92
Number of Members in the Household	5.93	4.48	5.81	6.33	7.21
Number of Adult Members in the Household	2.93	2.48	2.83	3.08	3.34
Household Has Phone Access (binary 0/1)	0.93	0.92	0.93	0.93	0.95
Household Has Internet Access (binary 0/1)	0.36	0.41	0.34	0.32	0.34
Probability of Moderate Food Insecurity	0.46	0.65	0.47	0.39	0.31
Food Consumption Score	49.02	48.01	48.20	48.92	50.14
Received Non-Agricultural Loan (binary 0/1)	0.09	0.10	0.09	0.07	0.07

[a] Winsorized to top 1%.

[b] Imputation performed by Evans School Policy Analysis and Research Group (UW EPAR, 2025).

[c] Characteristics of the Head of Household, coded verbatim from survey responses.

Note: We conducted ANOVA p-value tests of equality across quartile means and found all variables statistically significant (at least at the 10% level) except Income Shocks.

TABLE A4.
Wave 4 Summary Statistics by Farm Size Quartile

	Mean	Mean by Farm Size Quartiles			
	Full Sample	Q1	Q2	Q3	Q4
Agricultural Loan Variables					
Received Agricultural Loan (Binary 0/1)	0.05	0.05	0.04	0.05	0.07
Size of Agricultural Loan (1,000s ₦) (Recipients Only)	76.72	107.41	86.62	53.44	67.89
Outcome Variables					
Input Expenditures (1,000s ₦)[a]	26.40	10.29	19.24	32.89	48.97
Land Expenditures (1,000s ₦)[a]	4.57	2.91	4.03	4.84	6.90
Labor Expenditures (1,000s ₦)[a]	66.87	38.40	56.24	64.53	108.20
Livestock Expenditures (1,000s ₦)[a]	19.61	11.01	17.53	18.99	26.21
Farm Expenditures (1,000s ₦)	106.99	57.17	89.63	114.84	184.43
Food Expenditures (1,000s ₦)[a]	1,338.82	1,389.48	1,318.80	1,361.76	1,252.38
Non-Food Expenditures (1,000s ₦)[a]	287.87	295.75	264.42	267.50	305.17
Nonfarm Expenditures (1,000s ₦)	1,626.70	1,685.23	1,583.22	1,629.26	1,557.55
All Fertilizer Use (kg/ha)	872.35	1,039.48	1,264.23	926.26	451.47
Ratios					
Agricultural Loan to Total Farm Expenditures	71.7%	187.9%	96.6%	46.5%	36.8%
Agricultural Loan to Nonfarm Expenditures	4.7%	6.4%	5.5%	3.3%	4.4%
Control Variables					
Farming Land Area (Hectares)[a,b]	1.40	0.15	0.54	1.29	3.62
Value of Farm Production (1,000s ₦)[a,b]	338.65	79.93	205.46	366.27	702.95
Assets (1,000s ₦)[a,b]	193.05	209.40	155.39	180.41	213.76
Nonfarm Income (1,000s ₦)[a,b]	272.69	289.36	244.82	265.33	239.32
Livestock Holdings (#)[a,b]	0.93	0.23	0.55	0.95	1.89
Has Formal Land Rights (binary 0/1)[a,b]	0.17	0.15	0.16	0.17	0.20
Income Shock (#)	0.25	0.32	0.22	0.24	0.22
Production Shock (#)	0.38	0.41	0.36	0.39	0.39
Price Shock (#)	0.19	0.23	0.19	0.22	0.15
Head Marital Status (binary 0/1)[c]	0.78	0.62	0.77	0.86	0.91
Head Age[c]	50.02	53.17	49.92	48.18	48.59
Head Sex (binary 1 ≡ Male)[c]	0.83	0.66	0.83	0.89	0.95
Number of Members in the Household	5.91	4.67	5.57	6.29	7.26
Number of Adult Members in the Household	2.88	2.55	2.74	2.97	3.28
Household Has Phone Access (binary 0/1)	0.94	0.94	0.93	0.95	0.95
Household Has Internet Access (binary 0/1)	0.36	0.40	0.33	0.33	0.34
Probability of Moderate Food Insecurity	0.33	0.55	0.31	0.25	0.21
Food Consumption Score	51.43	51.31	50.25	51.22	52.23
Received Non-Agricultural Loan (binary 0/1)	0.10	0.13	0.10	0.08	0.07

[a] Winsorized to top 1%.

[b] Imputation performed by Evans School Policy Analysis and Research Group (UW EPAR, 2025).

[c] Characteristics of the Head of Household, coded verbatim from survey responses.

TABLE A5.
Wave 5 Summary Statistics by Farm Size Quartile

	Mean	Mean by Farm Size Quartiles			
	Full Sample	Q1	Q2	Q3	Q4
Agricultural Loan Variables					
Received Agricultural Loan (Binary 0/1)	0.05	0.03	0.05	0.06	0.06
Size of Agricultural Loan (1,000s ₦) (Recipients Only)	129.83	110.04	77.15	141.63	150.27
Outcome Variables					
Input Expenditures (1,000s ₦)[a]	40.01	15.76	35.95	49.25	67.06
Land Expenditures (1,000s ₦)[a]	4.02	2.36	4.54	4.11	5.64
Labor Expenditures (1,000s ₦)[a]	62.27	26.26	52.75	72.97	97.20
Livestock Expenditures (1,000s ₦)[a]	47.79	28.12	53.80	49.35	54.82
Farm Expenditures (1,000s ₦)	128.98	54.20	119.23	155.17	205.06
Food Expenditures (1,000s ₦)[a]	1,737.89	1,849.36	1,843.66	1,705.43	1,404.82
Non-Food Expenditures (1,000s ₦)[a]	567.46	558.70	540.48	563.52	577.78
Nonfarm Expenditures (1,000s ₦)	2,305.36	2,408.06	2,384.14	2,268.95	1,982.60
All Fertilizer Use (kg/ha)	633.75	788.85	991.52	602.96	275.97
Ratios					
Agricultural Loan to Total Farm Expenditures	100.7%	203.0%	64.7%	91.3%	73.3%
Agricultural Loan to Nonfarm Expenditures	5.6%	4.6%	3.2%	6.2%	7.6%
Control Variables					
Farming Land Area (Hectares)[a,b]	0.97	0.09	0.35	0.82	2.62
Value of Farm Production (1,000s ₦)[a,b]	372.75	76.53	265.03	463.71	686.60
Assets (1,000s ₦)[a,b]	360.31	417.63	310.93	349.51	348.07
Nonfarm Income (1,000s ₦)[a,b]	427.05	424.75	429.65	421.51	385.06
Livestock Holdings (#)[a,b]	0.65	0.19	0.42	0.67	1.18
Has Formal Land Rights (binary 0/1)[a,b]	0.12	0.09	0.09	0.13	0.17
Income Shock (#)	0.15	0.12	0.13	0.17	0.19
Production Shock (#)	1.52	1.43	1.52	1.54	1.68
Price Shock (#)	1.94	2.13	1.94	1.92	1.79
Head Marital Status (binary 0/1)[c]	0.76	0.58	0.80	0.81	0.86
Head Age[c]	53.47	57.16	52.66	51.40	51.99
Head Sex (binary 1 ≡ Male)[c]	0.79	0.63	0.83	0.84	0.88
Number of Members in the Household	5.94	4.24	5.89	6.58	7.18
Number of Adult Members in the Household	2.98	2.40	2.88	3.19	3.46
Household Has Phone Access (binary 0/1)	0.92	0.92	0.91	0.93	0.94
Household Has Internet Access (binary 0/1)	0.37	0.44	0.34	0.34	0.33
Probability of Moderate Food Insecurity	0.58	0.72	0.59	0.53	0.49
Food Consumption Score	46.61	45.84	46.39	46.79	46.54
Received Non-Agricultural Loan (binary 0/1)	0.07	0.08	0.08	0.06	0.07

[a] Winsorized to top 1%.

[b] Imputation performed by Evans School Policy Analysis and Research Group (UW EPAR, 2025).

[c] Characteristics of the Head of Household, coded verbatim from survey responses.

TABLE A6.
Heterogeneous Effects of Agricultural Loans on Disaggregated Farm Expenditures (TWFE)

	Input Exp. (log) (1)	Land Exp. (log) (2)	Labor Exp. (log) (3)	Livestock Exp. (log) (4)	Fertilizer (log kg/ha) (5)
Received Agricultural Loan (Loan)	0.099	0.422	−0.001	0.317	0.131
	(0.134)	(0.256)	(0.109)	(0.277)	(0.166)
Loan × Farming land area (HA)	0.094	0.083	0.086	0.072	−0.005
	(0.090)	(0.216)	(0.077)	(0.118)	(0.078)
Specification					
Controls	Yes	Yes	Yes	Yes	Yes
Federal State FEs	Yes	Yes	Yes	Yes	Yes
Survey Wave FEs	Yes	Yes	Yes	Yes	Yes
Household FEs	Yes	Yes	Yes	Yes	Yes
Adjusted-R^2	0.400	0.336	0.216	0.211	0.422
Within Adjusted-R^2	0.061	0.010	0.055	0.047	0.113
Observations	3,670	352	4,010	1,176	1,676

Standard errors in parentheses. Two-way fixed effects (TWFE) with cluster-robust standard errors at the household level. Loan is evaluated at mean farm size.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A7.
Heterogeneous Effects of Agricultural Loans on Nonfarm Expenditures (TWFE)

	Food Expenditures (log)	Non-Food Expenditures (log)
Received Agricultural Loan (Loan)	0.178**	0.084
	(0.085)	(0.052)
Loan × Farming land area (HA)	−0.045	0.012
	(0.049)	(0.036)
Specification		
Controls	Yes	Yes
Federal State FEs	Yes	Yes
Survey Wave FEs	Yes	Yes
Household FEs	Yes	Yes
Adjusted-R^2	0.350	0.602
Within Adjusted-R^2	0.069	0.126
Observations	5,898	5,900

Standard errors in parentheses. Two-way fixed effects (TWFE) with cluster-robust standard errors at the household level. Loan is evaluated at mean farm size.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A8.
Baseline Sensitivity Analysis for Farm and Nonfarm Expenditures

	3 Folds	5 Folds	8 Folds
Panel A: Log Farm Expenditures			
Received Agricultural Loan	0.137*	0.136*	0.135*
	(0.072)	(0.071)	(0.071)
R_Y^2 (Outcome: $Y \sim X$)	0.346	0.351	0.352
AUC (Loan)	0.619	0.626	0.622
AP (Loan)	0.083	0.083	0.082
Observations	5,476	5,476	5,476
Panel B: Log Nonfarm Expenditures			
Received Agricultural Loan	0.137***	0.135***	0.132***
	(0.049)	(0.049)	(0.048)
R_Y^2 (Outcome: $Y \sim X$)	0.417	0.420	0.422
AUC (Loan)	0.622	0.630	0.625
AP (Loan)	0.081	0.082	0.080
Observations	5,900	5,900	5,900
Specification			
Controls	Yes	Yes	Yes
Federal State FEs	Yes	Yes	Yes
Survey Wave FEs	Yes	Yes	Yes
Household CREs	Yes	Yes	Yes
Learners			
Loan $\sim X$	PLH-L	PLH-L	PLH-L
Expenditures $\sim X$	ERH-R	ERH-R	ERH-R

Standard errors in parentheses. Double Machine Learning (PLR), $n_{\text{rep}} = 30$.

R_Y^2 reports nuisance learner performance for the outcome model.

AUC (Area Under Curve) and AP (Average Precision) report nuisance learner performance for the treatment model.

ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.

PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked with a Logistic meta-learner.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A9.
Heterogeneous Sensitivity Analysis for Farm and Nonfarm Expenditures

	3 Folds	5 Folds	8 Folds
Panel A: Log Farm Expenditures			
Received Agricultural Loan	0.126*	0.124*	0.125*
	(0.073)	(0.072)	(0.072)
Loan × Farming Land Area (HA)	0.068	0.067	0.067
	(0.046)	(0.045)	(0.045)
R^2_Y (Outcome: $Y \sim X$)	0.346	0.351	0.352
AUC (Loan)	0.619	0.626	0.622
AP (Loan)	0.083	0.083	0.082
Observations	5,476	5,476	5,476
Panel B: Log Nonfarm Expenditures			
Received Agricultural Loan	0.142***	0.139***	0.137***
	(0.049)	(0.049)	(0.048)
Loan × Farming Land Area (HA)	−0.033	−0.031	−0.031
	(0.031)	(0.031)	(0.031)
R^2_Y (Outcome: $Y \sim X$)	0.417	0.420	0.422
AUC (Loan)	0.622	0.630	0.625
AP (Loan)	0.081	0.082	0.080
Observations	5,900	5,900	5,900
Specification			
Controls	Yes	Yes	Yes
Federal State FEs	Yes	Yes	Yes
Survey Wave FEs	Yes	Yes	Yes
Household CREs	Yes	Yes	Yes
Learners			
Loan $\sim X$	PLH-L	PLH-L	PLH-L
Expenditures $\sim X$	ERH-R	ERH-R	ERH-R

Standard errors in parentheses. Double Machine Learning (PLR), $n_{\text{rep}} = 30$.

R^2_Y reports nuisance learner performance for the outcome model.

AUC (Area Under Curve) and AP (Average Precision) report nuisance learner performance for the treatment model.

ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.

PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked with a Logistic meta-learner.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A10.
Average Treatment Effects of Any Loan on Expenditures

	Farm Expenditures (log)		Nonfarm Expenditures (log)	
	DML	TWFE	DML	TWFE
Received Any Loan	0.057	0.052	0.050	0.073
	(0.047)	(0.067)	(0.032)	(0.045)
Specification				
Controls	Yes	Yes	Yes	Yes
Federal State FEs	Yes	Yes	Yes	Yes
Survey Wave FEs	Yes	Yes	Yes	Yes
Household CREs	Yes	No	Yes	No
Household FEs	No	Yes	No	Yes
Learners				
Loan $\sim X$	PLH-L		PLH-L	
Expenditures $\sim X$	ERH-R		ERH-R	
K-folds	5		5	
Diagnostics				
R^2_Y (Outcome: $Y \sim X$)	0.351		0.421	
AUC (Loan $\sim X$)	0.680		0.681	
AP (Loan $\sim X$)	0.212		0.213	
Adjusted-R^2		0.353		0.428
Within Adjusted-R^2		0.074		0.088
Observations	5,476	5,476	5,900	5,900

Standard errors in parentheses. Columns 1 and 3 are Double Machine Learning (PLR), $n_{\text{rep}} = 30$; columns 2 and 4 are TWFE with cluster-robust standard errors at the household level.
Adj-R^2 and Within Adj-R^2 report TWFE model fit.
R^2_Y reports nuisance learner performance for the outcome model.
AUC (Area Under Curve) and AP (Average Precision) report nuisance learner performance for the treatment model.
ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.
PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked with a Logistic meta-learner.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A11.
Heterogeneous Effects of Any Loan on Expenditures

	Farm Expenditures (log)		Nonfarm Expenditures (log)	
	DML	TWFE	DML	TWFE
Received Any Loan (Loan)	0.058	0.050	0.051	0.073
	(0.047)	(0.067)	(0.032)	(0.045)
Loan × Farming Land Area (HA)	0.031	0.095**	−0.001	0.001
	(0.032)	(0.047)	(0.023)	(0.029)
Specification				
Controls	Yes	Yes	Yes	Yes
Federal State FEs	Yes	Yes	Yes	Yes
Survey Wave FEs	Yes	Yes	Yes	Yes
Household CREs	Yes	No	Yes	No
Household FEs	No	Yes	No	Yes
Learners				
Loan $\sim X$	PLH-L		PLH-L	
Expenditures $\sim X$	ERH-R		ERH-R	
K-folds	5		5	
Diagnostics				
R^2_Y (Outcome: $Y \sim X$)	0.351		0.421	
AUC (Loan $\sim X$)	0.680		0.681	
AP (Loan $\sim X$)	0.212		0.213	
Adjusted-R^2		0.353		0.428
Within Adjusted-R^2		0.076		0.088
Observations	5,476	5,476	5,900	5,900

Standard errors in parentheses. Columns 1 and 3 are Double Machine Learning (PLR), $n_{\text{rep}} = 30$;
columns 2 and 4 are TWFE with cluster-robust standard errors at the household level.
Adj-R^2 and Within Adj-R^2 report TWFE model fit.
R^2_Y reports nuisance learner performance for the outcome model.
AUC (Area Under Curve) and AP (Average Precision) report nuisance learner performance for the treatment model.
ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.
PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked with a Logistic meta-learner.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A12.
Heterogeneous Effects of Any Loan on Disaggregated Farm Expenditures

	Input Exp. (log) (1)	Land Exp. (log) (2)	Labor Exp. (log) (3)	Livestock Exp. (log) (4)	Fertilizer (log kg/ha) (5)
Received Any Loan (Loan)	−0.037	−0.015	−0.002	0.025	0.134
	(0.065)	(0.135)	(0.054)	(0.144)	(0.095)
Loan × Farming land area (HA)	−0.021	0.011	0.004	0.054	0.142**
	(0.046)	(0.133)	(0.036)	(0.074)	(0.060)
Specification					
Controls	Yes	Yes	Yes	Yes	Yes
Federal State FEs	Yes	Yes	Yes	Yes	Yes
Survey Wave FEs	Yes	Yes	Yes	Yes	Yes
Household CREs	Yes	Yes	Yes	Yes	Yes
Learners					
Loan $\sim X$	PLH-L	PLH-L	PLH-L	PLH-L	PLH-L
Expenditures $\sim X$	ERH-R	ERH-R	ERH-R	ERH-R	ERH-R
K-folds	5	5	5	5	5
Diagnostics					
R^2_Y (Outcome: $Y \sim X$)	0.362	0.241	0.238	0.224	0.385
AUC (Loan $\sim X$)	0.683	0.580	0.685	0.696	0.642
AP (Loan $\sim X$)	0.214	0.237	0.214	0.229	0.208
Observations	3,670	352	4,010	1,176	1,676

Standard errors in parentheses. Double Machine Learning (PLR), $n_{\text{rep}} = 30$.

R^2_Y reports nuisance learner performance for the outcome model.

AUC (Area Under Curve) and AP (Average Precision) report nuisance learner performance for the treatment model.

ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.

PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked with a Logistic meta-learner.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A13.
Heterogeneous Effects of Any Loan on Nonfarm Expenditures

	Food Expenditures (log)	Non-Food Expenditures (log)
Received Any Loan (Loan)	0.045	0.061***
	(0.039)	(0.023)
Loan × Farming land area (HA)	−0.017	0.024
	(0.028)	(0.017)
Specification		
Controls	Yes	Yes
Federal State FEs	Yes	Yes
Survey Wave FEs	Yes	Yes
Household CREs	Yes	Yes
Learners		
Loan $\sim X$	PLH-L	PLH-L
Expenditures $\sim X$	ERH-R	ERH-R
K-folds	5	5
Diagnostics		
R^2_Y (Outcome: $Y \sim X$)	0.336	0.604
AUC (Loan $\sim X$)	0.681	0.681
AP (Loan $\sim X$)	0.213	0.213
Observations	5,898	5,900

Standard errors in parentheses. Double Machine Learning (PLR), $n_{\text{rep}} = 30$.

R^2_Y reports nuisance learner performance for the outcome model.

AUC (Area Under Curve) and AP (Average Precision) report nuisance learner performance for the treatment model.

ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.

PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked with a Logistic meta-learner.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A14.
Heterogeneous Effects of Any Loan on Disaggregated Farm Expenditures (TWFE)

	Input Exp. (log) (1)	Land Exp. (log) (2)	Labor Exp. (log) (3)	Livestock Exp. (log) (4)	Fertilizer (log kg/ha) (5)
Received Any Loan (Loan)	0.010	0.222	−0.055	0.073	0.185
	(0.087)	(0.223)	(0.071)	(0.198)	(0.139)
Loan × Farming land area (HA)	0.008	0.207	0.017	0.175*	−0.023
	(0.059)	(0.212)	(0.054)	(0.101)	(0.091)
Specification					
Controls	Yes	Yes	Yes	Yes	Yes
Federal State FEs	Yes	Yes	Yes	Yes	Yes
Survey Wave FEs	Yes	Yes	Yes	Yes	Yes
Household FEs	Yes	Yes	Yes	Yes	Yes
Adjusted-R^2	0.400	0.339	0.215	0.214	0.422
Within Adjusted-R^2	0.060	0.014	0.054	0.051	0.113
Observations	3,670	352	4,010	1,176	1,676

Standard errors in parentheses. Two-way fixed effects (TWFE) with cluster-robust standard errors at the household level. Loan is evaluated at mean farm size.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A15.
Heterogeneous Effects of Any Loan on Nonfarm Expenditures (TWFE)

	Food Expenditures (log)	Non-Food Expenditures (log)
Received Any Loan (Loan)	0.073	0.068**
	(0.057)	(0.033)
Loan × Farming land area (HA)	0.000	−0.005
	(0.037)	(0.023)
Specification		
Controls	Yes	Yes
Federal State FEs	Yes	Yes
Survey Wave FEs	Yes	Yes
Household FEs	Yes	Yes
Adjusted-R^2	0.349	0.602
Within Adjusted-R^2	0.068	0.126
Observations	5,898	5,900

Standard errors in parentheses. Two-way fixed effects (TWFE) with cluster-robust standard errors at the household level. Loan is evaluated at mean farm size.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A16.
Balanced-Sample Farm and Nonfarm Expenditure Effects (DML)

	Farm Expenditures (log)	Nonfarm Expenditures (log)
Received Agricultural Loan (Loan)	0.125^{*}	0.141^{***}
	(0.072)	(0.050)
Loan $\times$ Farming land area (HA)	0.068	-0.031
	(0.045)	(0.031)
Specification		
Controls	Yes	Yes
Federal State FEs	Yes	Yes
Survey Wave FEs	Yes	Yes
Household CREs	Yes	Yes
Learners		
Loan $\sim X$	PLH-L	PLH-L
Expenditures $\sim X$	ERH-R	ERH-R
K-folds	5	5
Diagnostics		
R^2_Y (Outcome model)	0.351	0.412
AUC (Loan $\sim X$)	0.575	0.575
AP (Loan $\sim X$)	0.069	0.069
Observations	5,476	5,476

Standard errors in parentheses. Double Machine Learning (PLR), $n_{\text{rep}} = 30$. Loan is evaluated at mean farm size.

Both equations use the same balanced second-stage sample of 5,476 observations (2,738 households); the household-clustered joint regression retains cross-outcome covariance.

R^2_Y reports nuisance learner performance for the outcome model. AUC (Area Under Curve) and AP (Average Precision) report nuisance learner performance for the treatment model.

ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner. PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked with a Logistic meta-learner.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A17.
Average Treatment Effects of Agricultural Extension on Expenditures

	Farm Expenditures (log)		Nonfarm Expenditures (log)	
	DML	TWFE	DML	TWFE
Panel A: Received Extension (PP)				
Received Extension	0.121***	0.146**	0.033	0.027
	(0.039)	(0.057)	(0.029)	(0.039)
Diagnostics				
R^2_Y (Outcome: $Y \sim X$)	0.350		0.420	
AUC (Extension $\sim X$)	0.745		0.740	
AP (Extension $\sim X$)	0.426		0.416	
Adjusted-R^2		0.354		0.428
Within Adjusted-R^2		0.076		0.087
Panel B: Received Extension (PP or PH)				
Received Extension	0.068**	0.131***	0.040*	0.062*
	(0.032)	(0.046)	(0.023)	(0.032)
Diagnostics				
R^2_Y (Outcome: $Y \sim X$)	0.351		0.420	
AUC (Extension $\sim X$)	0.744		0.742	
AP (Extension $\sim X$)	0.625		0.615	
Adjusted-R^2		0.354		0.428
Within Adjusted-R^2		0.076		0.088
Specification				
Controls	Yes	Yes	Yes	Yes
Federal State FEs	Yes	Yes	Yes	Yes
Survey Wave FEs	Yes	Yes	Yes	Yes
Household CREs	Yes	No	Yes	No
Household FEs	No	Yes	No	Yes
Learners				
Extension $\sim X$	PLH-L		PLH-L	
Expenditures $\sim X$	ERH-R		ERH-R	
K-folds	5		5	
Observations	5,476	5,476	5,900	5,900

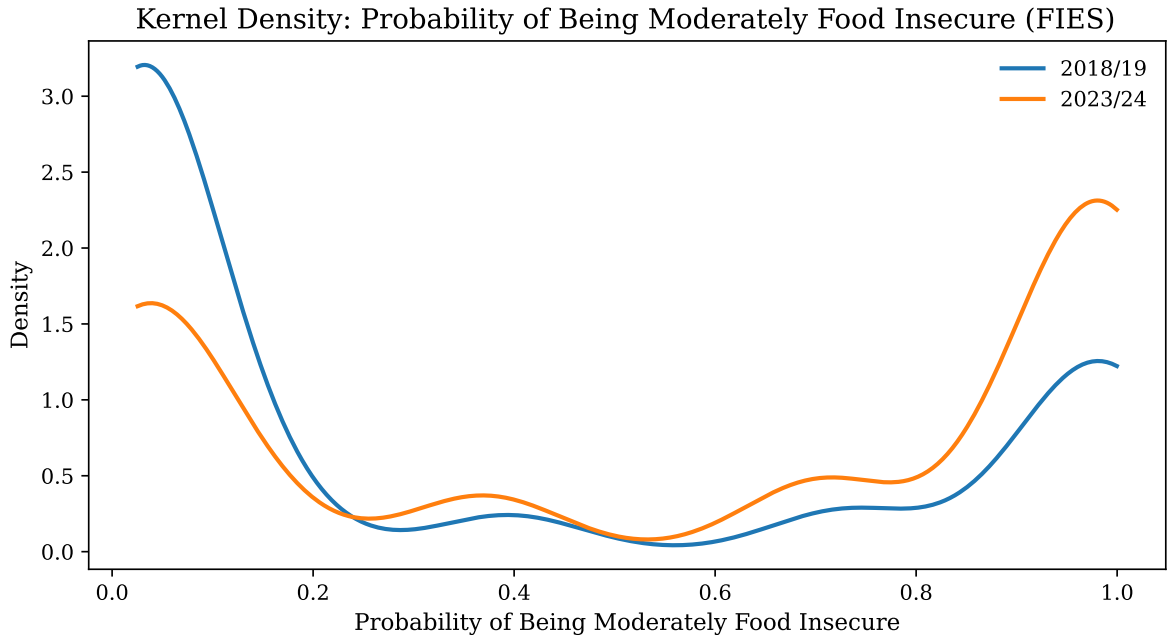
Standard errors in parentheses. Columns 1 and 3 are Double Machine Learning (PLR), $n_{\text{rep}} = 30$; columns 2 and 4 are TWFE with cluster-robust standard errors at the household level.
Adj-R^2 and Within Adj-R^2 report TWFE model fit. State indicators are absorbed by household FEs.
R^2_Y reports nuisance learner performance for the outcome model.
AUC (Area Under Curve) and AP (Average Precision) report nuisance learner performance for the treatment model. Diagnostics use predictions averaged across repetitions.
ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.
PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked with a Logistic meta-learner.
PP is post-planting; PP or PH is either survey visit. Each definition is estimated separately.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A18.
Heterogeneous Effects of Agricultural Extension on Expenditures

	Farm Expenditures (log)		Nonfarm Expenditures (log)	
	DML	TWFE	DML	TWFE
Panel A: Received Extension (PP)				
Extension at mean size	0.108***	0.177***	0.029	0.031
	(0.042)	(0.059)	(0.030)	(0.040)
Extension × Farming Land Area (HA)	0.009	−0.105***	0.005	−0.014
	(0.023)	(0.033)	(0.021)	(0.022)
Diagnostics				
R^2_Y (Outcome: $Y \sim X$)	0.350		0.420	
AUC (Extension $\sim X$)	0.745		0.740	
AP (Extension $\sim X$)	0.426		0.416	
Adjusted-R^2		0.356		0.428
Within Adjusted-R^2		0.079		0.087
Panel B: Received Extension (PP or PH)				
Extension at mean size	0.060*	0.138***	0.039*	0.066**
	(0.035)	(0.047)	(0.023)	(0.032)
Extension × Farming Land Area (HA)	0.037*	−0.053*	0.002	−0.022
	(0.021)	(0.030)	(0.017)	(0.019)
Diagnostics				
R^2_Y (Outcome: $Y \sim X$)	0.351		0.420	
AUC (Extension $\sim X$)	0.744		0.742	
AP (Extension $\sim X$)	0.625		0.615	
Adjusted-R^2		0.355		0.428
Within Adjusted-R^2		0.077		0.088
Specification				
Controls	Yes	Yes	Yes	Yes
Federal State FEs	Yes	Yes	Yes	Yes
Survey Wave FEs	Yes	Yes	Yes	Yes
Household CREs	Yes	No	Yes	No
Household FEs	No	Yes	No	Yes
Learners				
Extension $\sim X$	PLH-L		PLH-L	
Expenditures $\sim X$	ERH-R		ERH-R	
K-folds	5		5	
Observations	5,476	5,476	5,900	5,900

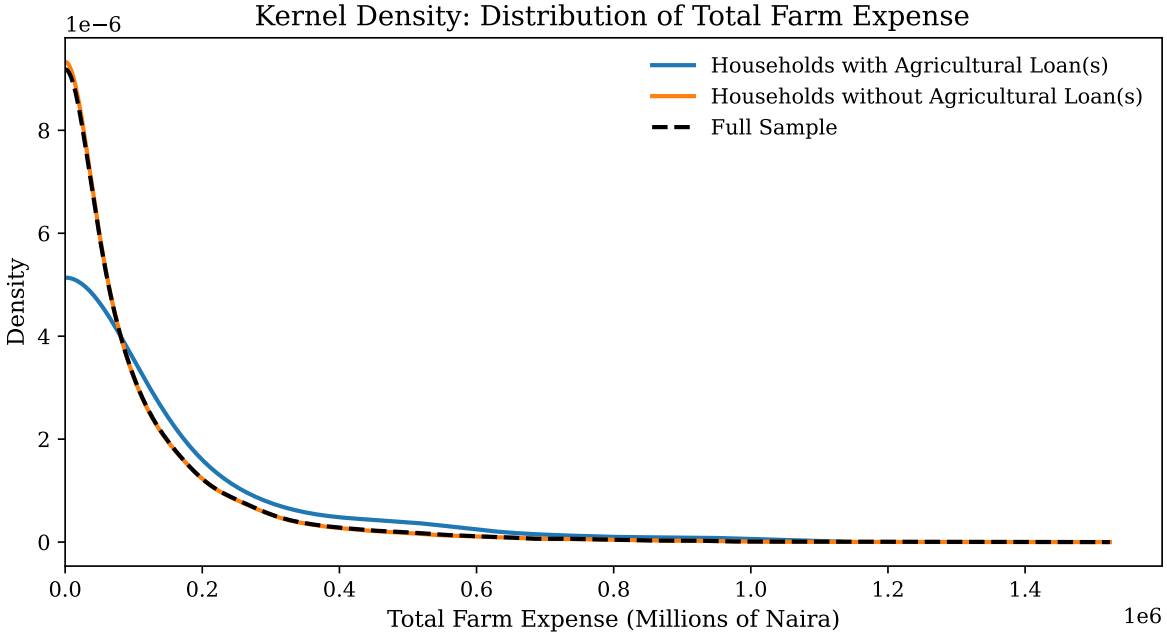
Standard errors in parentheses. Columns 1 and 3 are Double Machine Learning (PLR), $n_{\text{rep}} = 30$;
columns 2 and 4 are TWFE with cluster-robust standard errors at the household level.
Adj-R^2 and Within Adj-R^2 report TWFE model fit. State indicators are absorbed by household FEs.
R^2_Y reports nuisance learner performance for the outcome model.
AUC (Area Under Curve) and AP (Average Precision) report nuisance learner performance for the treatment model.
ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.
PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked with a Logistic meta-learner.
PP is post-planting; PP or PH is either survey visit. Each definition is estimated separately.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Descriptive Figures



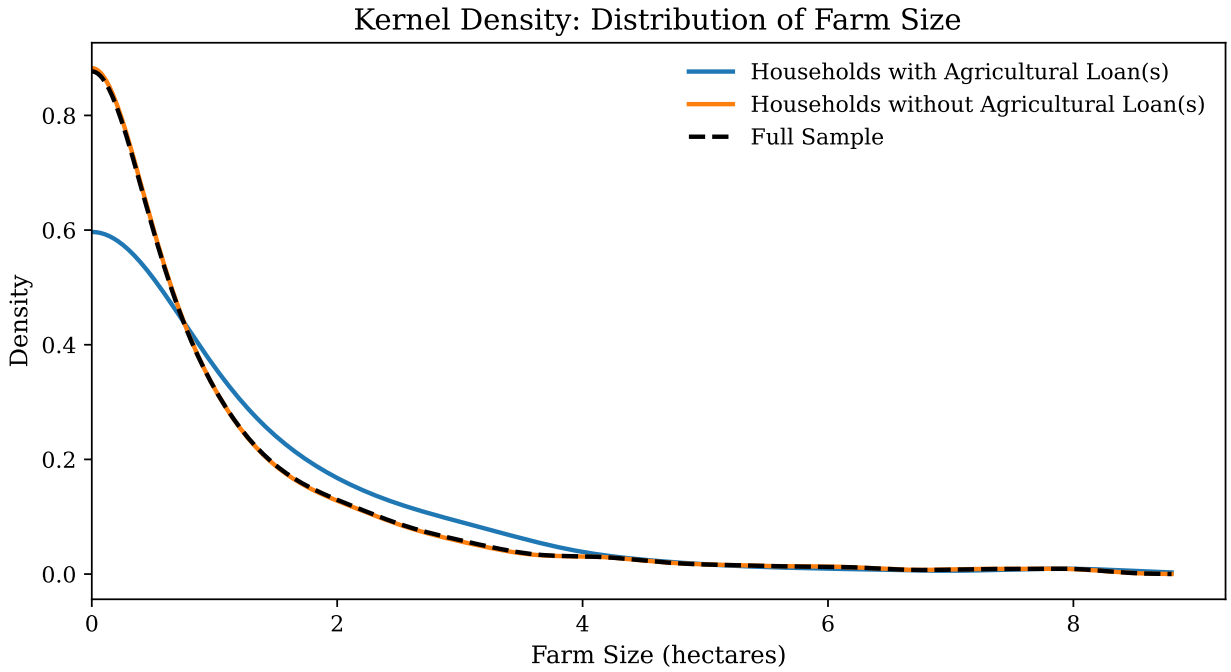
Notes: Kernel density of the probability of moderate food insecurity (FIES), by survey wave. Sample restricted to households observed in both waves.

FIGURE A1



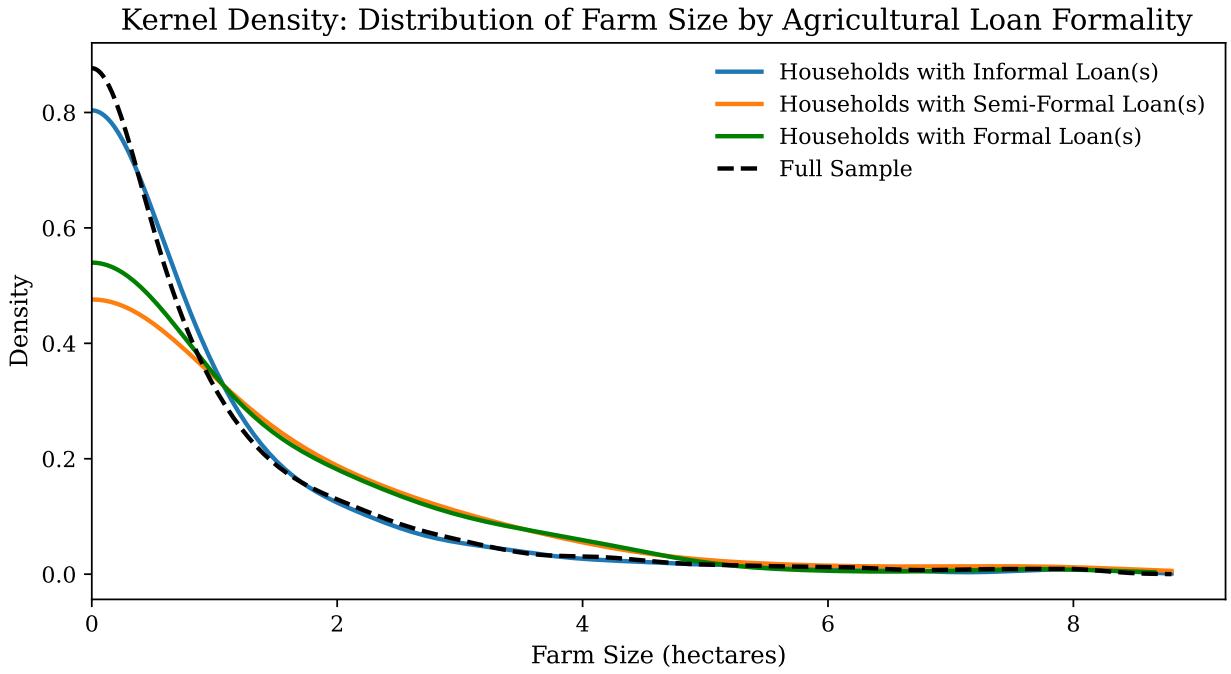
Notes: Kernel density of farm expenditures by agricultural loan receipt.

FIGURE A2



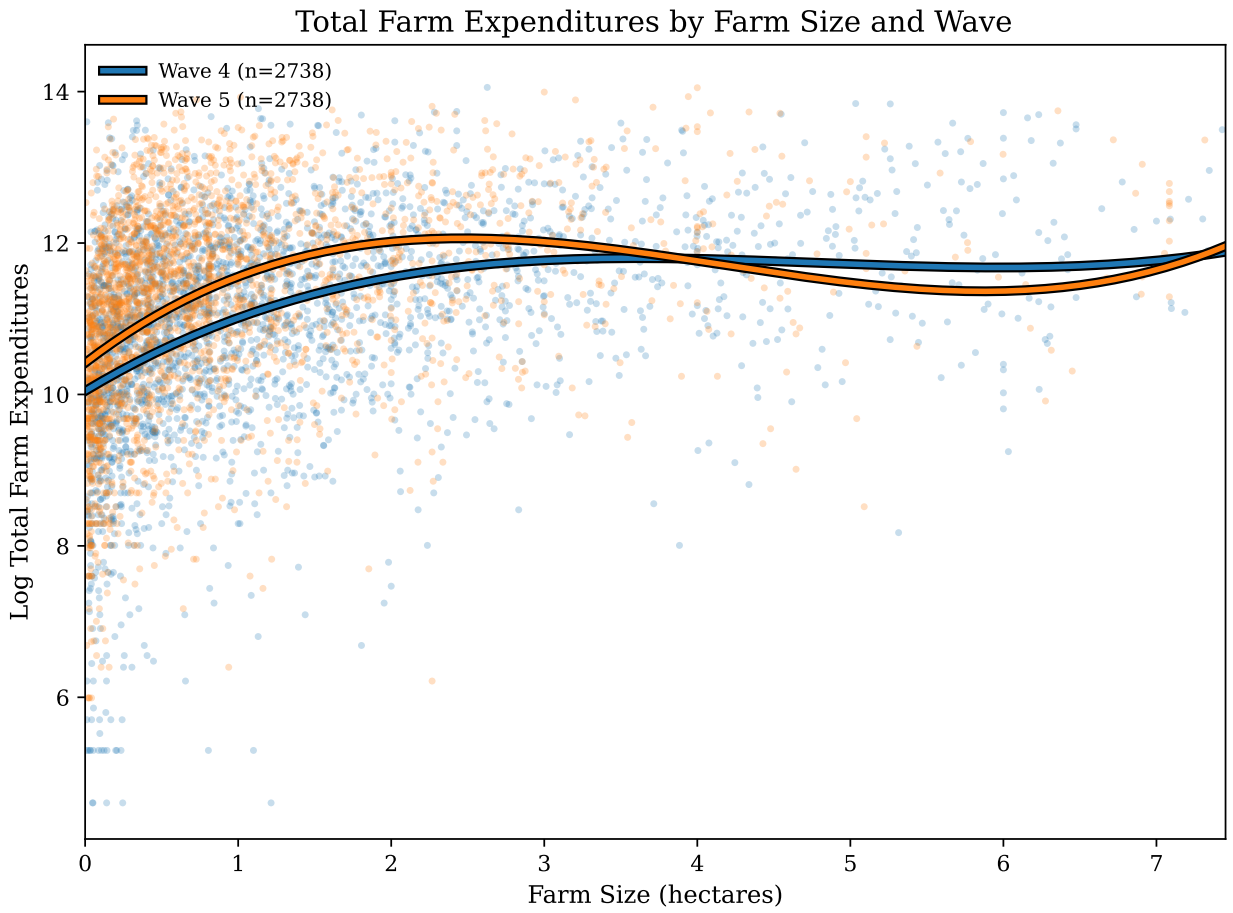
Notes: Kernel density of farm size by agricultural loan receipt.

FIGURE A3



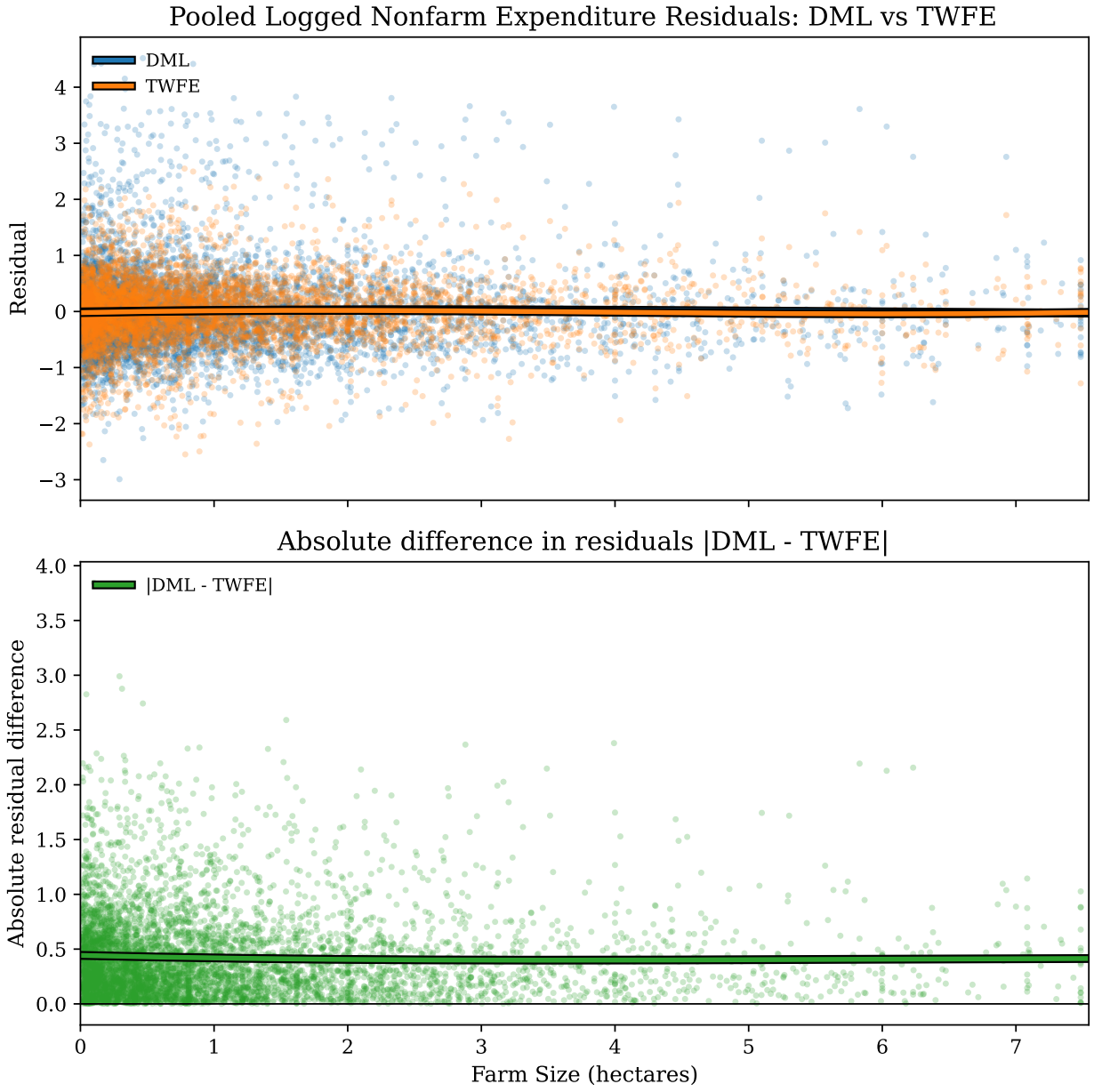
Notes: Kernel density of farm size by agricultural loan formality.

FIGURE A4



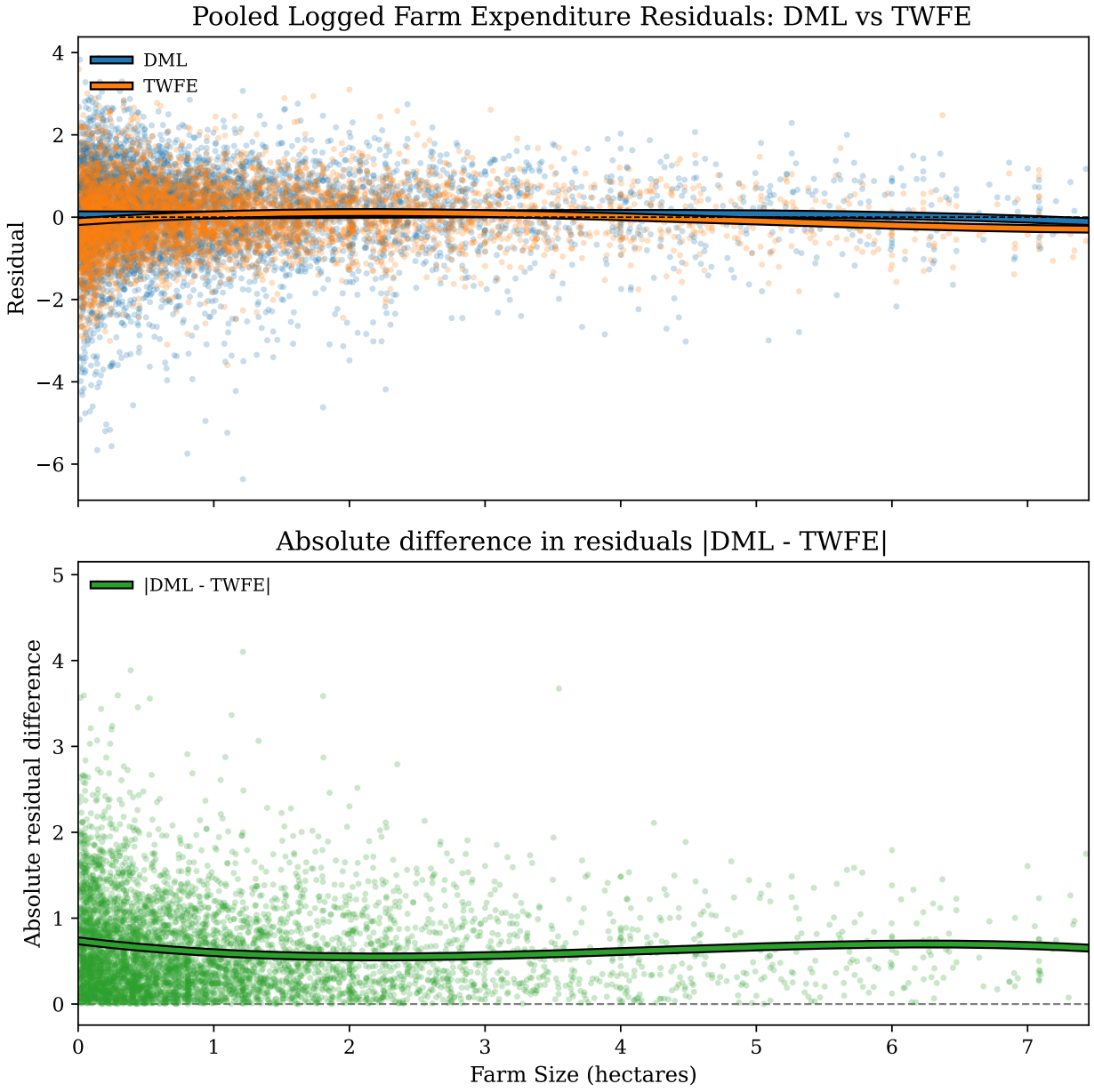
Notes: Log total farm expenditures by farm size and survey wave, with third-degree polynomial fits.

FIGURE A5



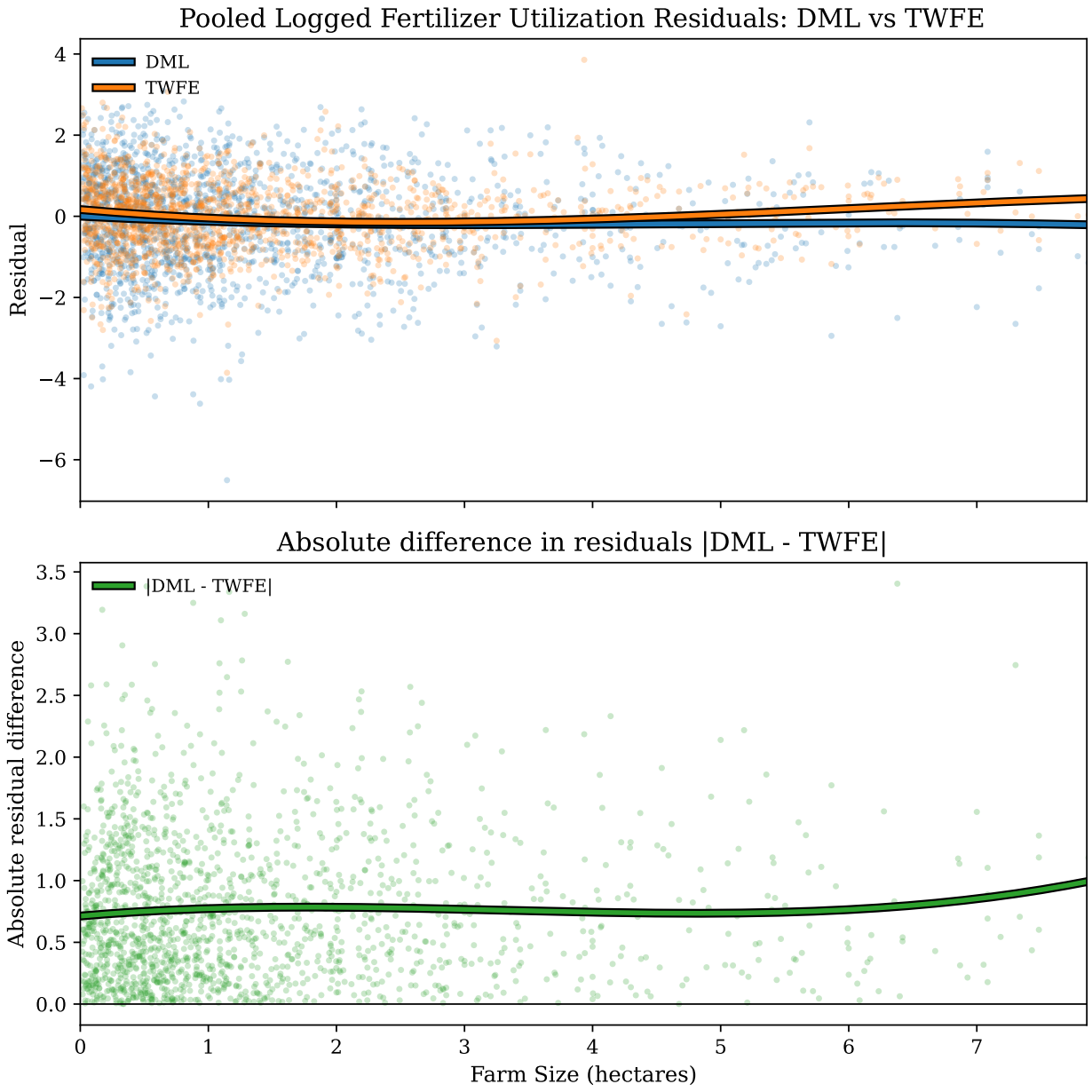
Notes: Top panel: DML residuals are out-of-fold nuisance residuals, $\tilde{Y}_{it} = Y_{it} - \hat{n}(X_{it}, \bar{X}_i)$; TWFE residuals are from an analogous specification estimated without the treatment terms, $\tilde{Y}_{it} = Y_{it} - (\hat{\alpha}_i + \hat{\lambda}_t + X'_{it}\hat{\gamma})$. Solid curves are third-degree polynomial fits. Bottom panel: absolute observation-level difference between the two residual series.

FIGURE A6



Notes: Top panel: DML residuals are out-of-fold nuisance residuals, $\tilde{Y}_{it} = Y_{it} - \hat{n}(X_{it}, \bar{X}_i)$; TWFE residuals are from an analogous specification estimated without the treatment terms, $\tilde{Y}_{it} = Y_{it} - (\hat{\alpha}_i + \hat{\lambda}_t + X'_{it}\hat{\gamma})$. Solid curves are third-degree polynomial fits. Bottom panel: absolute observation-level difference between the two residual series.

FIGURE A7



Notes: Top panel: DML residuals are out-of-fold nuisance residuals, $\tilde{Y}_{it} = Y_{it} - \hat{n}(X_{it}, \bar{X}_i)$; TWFE residuals are from an analogous specification estimated without the treatment terms, $\tilde{Y}_{it} = Y_{it} - (\hat{\alpha}_i + \hat{\lambda}_t + X'_{it}\hat{\gamma})$. Solid curves are third-degree polynomial fits. Bottom panel: absolute observation-level difference between the two residual series.

FIGURE A8

Appendix B

The GATE estimates in Table B1 broadly align with the linear interaction reported in Table 3, though with less precision given the further dilution of the already rare treatment. In Panel B, we observe that the impact of agricultural credit on farm expenditures among the smallest landholders (Q1) is small and statistically insignificant (0.031) while the effect on the largest landholders (Q4) is large and significant at the 5 percent level (0.235). In contrast, Panel A shows a large effect of agricultural credit on nonfarm expenditures among the smallest landholders significant at the 10 percent level (0.207) with some attenuation among the largest landholders (0.112). The endpoints of the farm size distribution are therefore generally consistent with the pattern observed in our continuous specification, in which credit is highly fungible, relaxing consumption constraints among all smallholders while additionally supporting investment among larger smallholders.

The middle quartiles (Q2 and Q3), however, present results indicating a non-monotonic pattern that the continuous specification cannot capture. We attribute this partially to the loss of power: treatment effects are now identified with between 58 and 93 treated observations per quartile. Additionally, an unexpected pattern emerges between Panels A and B which is worth noting. Across quartiles, it appears treatment effects are inversely correlated, both in statistical significance and magnitude of point estimates. It should be noted that the farm and nonfarm expenditure variables are constructed independently, and as such, this result is not a mechanical artifact of the data cleaning process. Speculatively, the inverse relationship might point directly to the fungibility of agricultural credit, reinforcing the expenditure reallocation interpretation developed in Section 4. That said, with the loss of precision in the quartile analysis, these results are ultimately inconclusive but present an opportunity for future research.

Table B1.
Group Average Treatment Effects (GATEs)
by Farm Size Quartile for Farm and Nonfarm Expenditures

	Q1	Q2	Q3	Q4
Panel A: Log Nonfarm Expenditures				
Received Agricultural Loan	0.207*	0.048	0.176*	0.112
	(0.106)	(0.086)	(0.094)	(0.100)
Treated	60	66	77	93
Observations	1,475	1,475	1,475	1,475
Panel B: Log Farm Expenditures				
Received Agricultural Loan	0.031	0.297**	−0.019	0.235**
	(0.167)	(0.149)	(0.144)	(0.109)
Treated	58	59	77	87
Observations	1,369	1,369	1,369	1,369
Specification				
Controls	Yes	Yes	Yes	Yes
Federal State FEs	Yes	Yes	Yes	Yes
Survey Wave FEs	Yes	Yes	Yes	Yes
Household CREs	Yes	Yes	Yes	Yes
Learners				
Loan $\sim X$	PLH-L	PLH-L	PLH-L	PLH-L
Expenditures $\sim X$	ERH-R	ERH-R	ERH-R	ERH-R
K-folds	5	5	5	5

Standard errors in parentheses. Double Machine Learning (PLR), $n_{\text{rep}} = 30$.

ERH-R: Elastic Net, Random Forest, and HistGradientBoosting Regressor with a Ridge meta-learner.

PLH-L: Poly/L1-Logit and HistGradientBoosting Classifier stacked with a Logistic meta-learner.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.